Approximate Matching

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Read

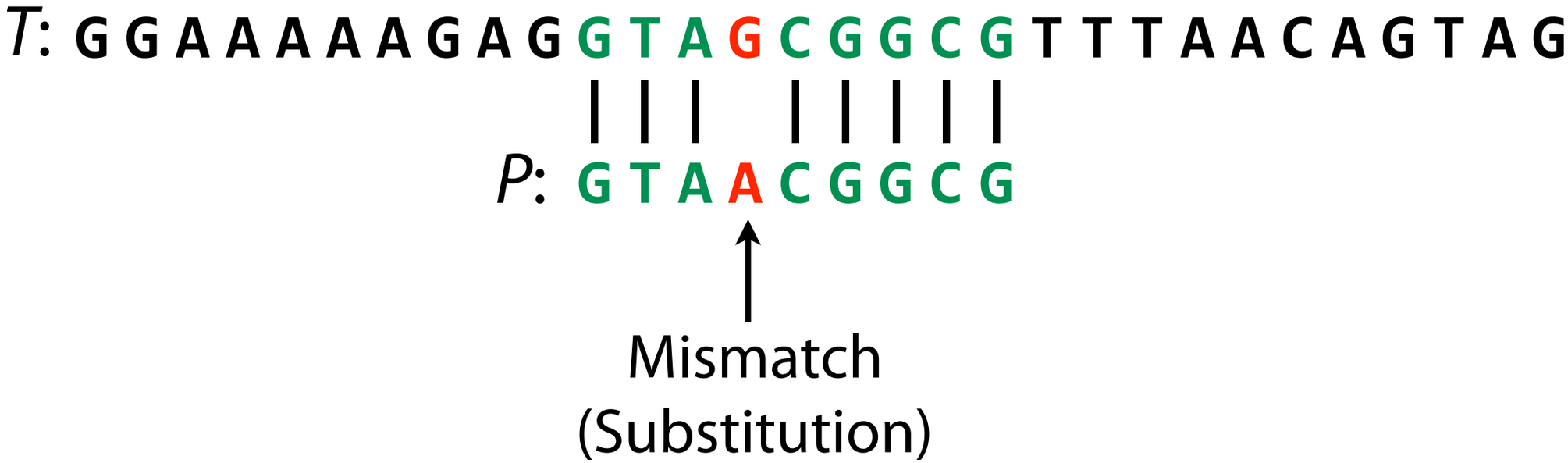
CTCAAACTCCTGACCTTTGGTGATCCACCCGCCTNGGCCTTC

Reference

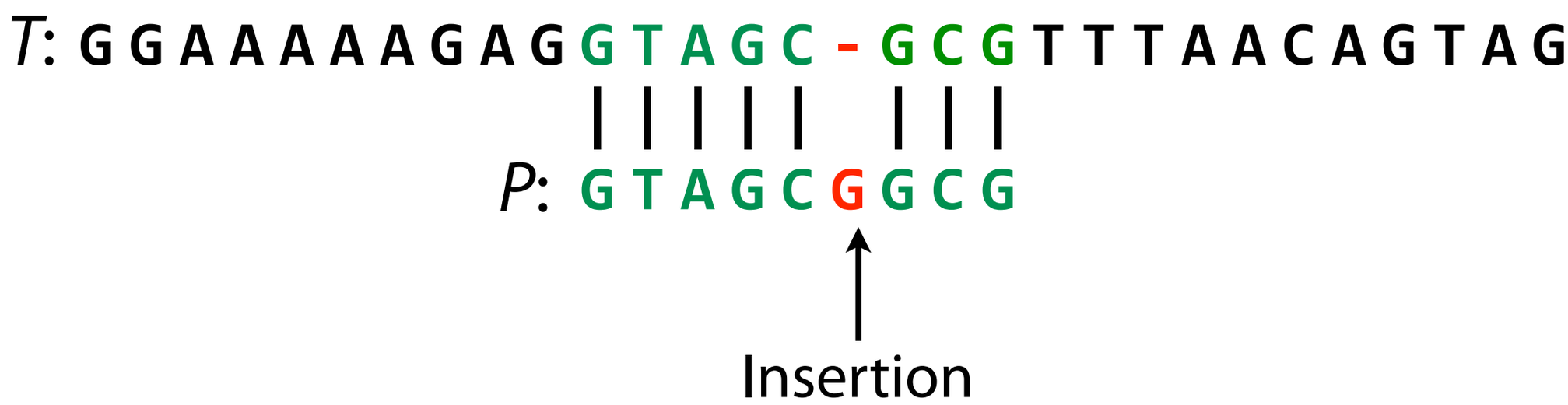
Sequence differences occur because of...

1. Sequencing error
2. Genetic variation

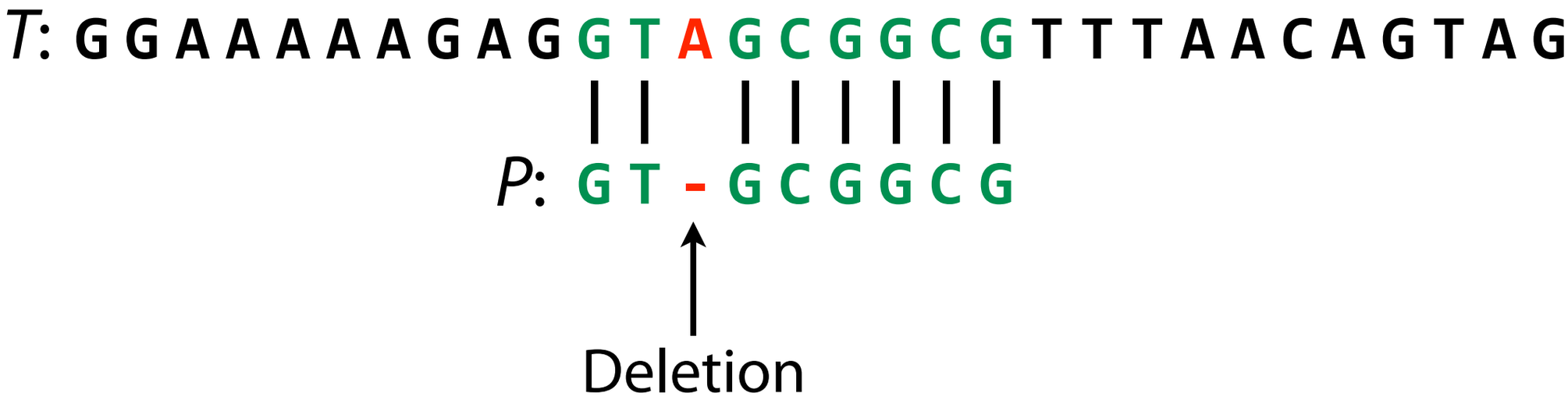
Approximate matching



Approximate matching



Approximate matching



Hamming distance

For X & Y where $|X| = |Y|$, *hamming distance* =
minimum # substitutions needed to turn one into the other

Hamming distance = 3

Edit distance

(AKA Levenshtein distance)

For *X* & *Y*, *edit distance* = minimum # edits (substitutions, insertions, deletions) needed to turn one into the other

```
X: T G G C C G C G C A A A A A C A G C
   | |   | | | | | | | | | |   | | | |
Y: T G A C C G C G C A A A A - C A G C
```

Edit distance = 2

```
X: G C G T A T G C G G C T A - A C G C
   | |   | | | | | | | | | |   | | |
Y: G C - T A T G C G G C T A T A C G C
```

Edit distance = 2

Approximate matching

Like exact matching, but *pattern P* may be within a certain *distance* (usually Hamming or edit) of *T*. Each such place is an *approximate match*.

Allowing edits is more challenging than just allowing mismatches

We'll return to edits

Approximate matching

```
def naive(p, t):
    occurrences = []
    for i in range(len(t) - len(p) + 1):  # loop over alignments
        match = True
        for j in range(len(p)):           # loop over characters
            if t[i+j] != p[j]:            # compare characters
                match = False             # mismatch; reject alignment
                break
        if match:
          occurrences.append(i)           # all chars matched; record
    return occurrences
```

Approximate matching

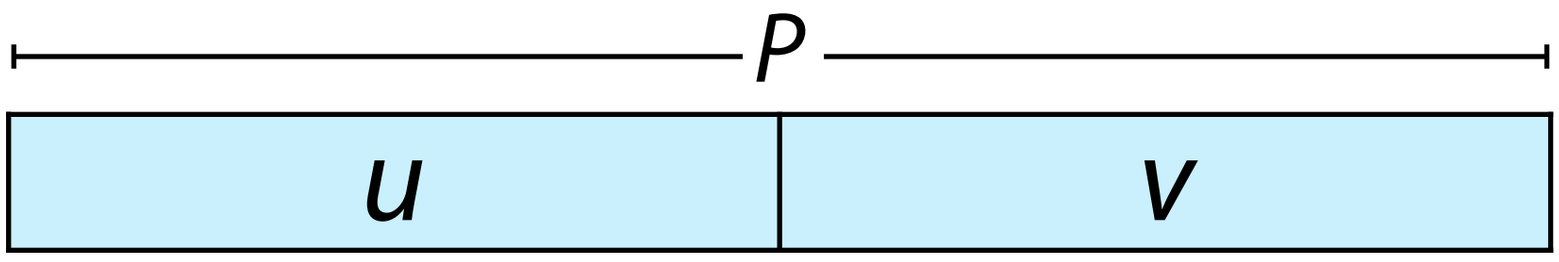
```
def naive_approx_hamming(p, t, maxDistance):
    occurrences = []
    for i in range(len(t) - len(p) + 1):  # loop over alignments
        nmm = 0
        for j in range(len(p)):           # loop over characters
            if t[i+j] != p[j]:            # compare characters
                nmm += 1                  # mismatch
                if nmm > maxDistance:
                    break                 # exceeded max hamming dist
    if nmm <= maxDistance:
      occurrences.append(i)               # approximate match
return occurrences
```

http://bit.ly/CG_NaiveApprox

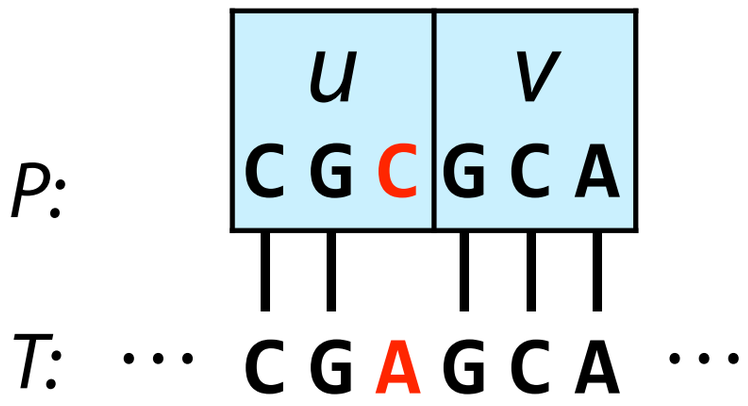
Approximate matching

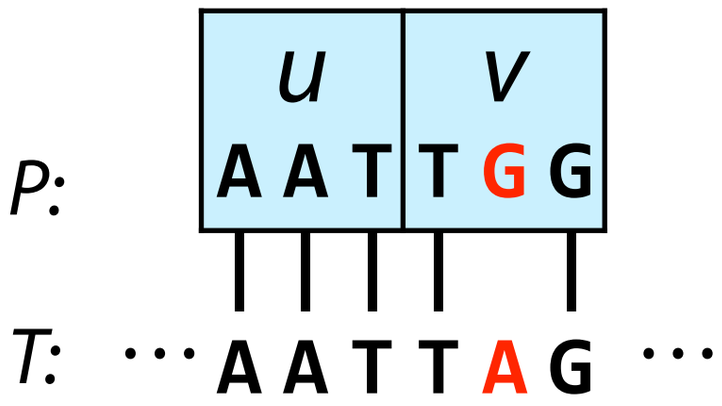
Wanted: way to apply exact matching algorithms to approximate matching problems

Approximate matching

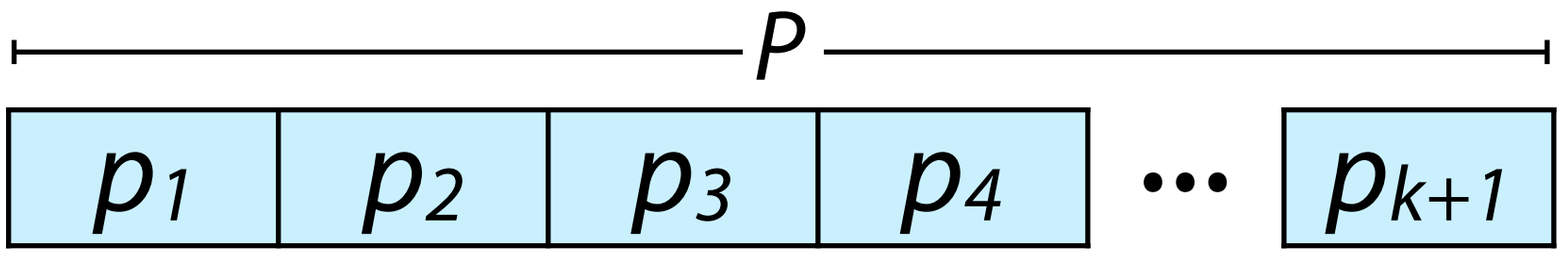


If *P* occurs in *T* with 1 edit, then *u* or *v* appears with no edits



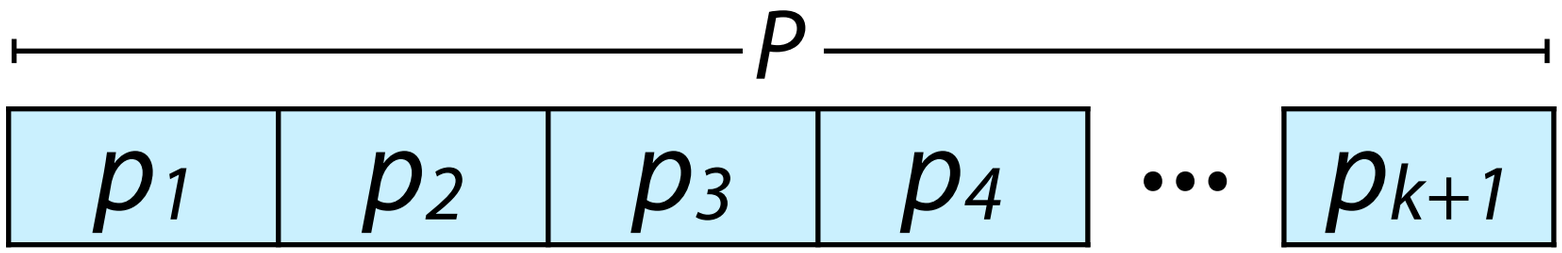


Approximate matching



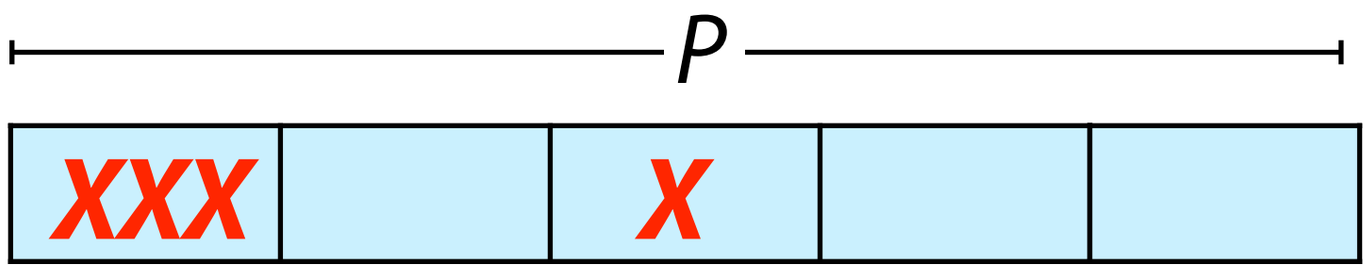
If *P* occurs in *T* with up to *k* edits…

Approximate matching

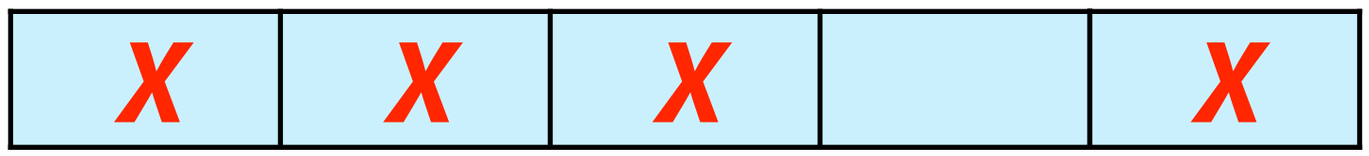


If *P* occurs in *T* with up to *k* edits, then at least one of $p_1, p_2, ..., p_{k+1}$ must appear with 0 edits

Approximate matching



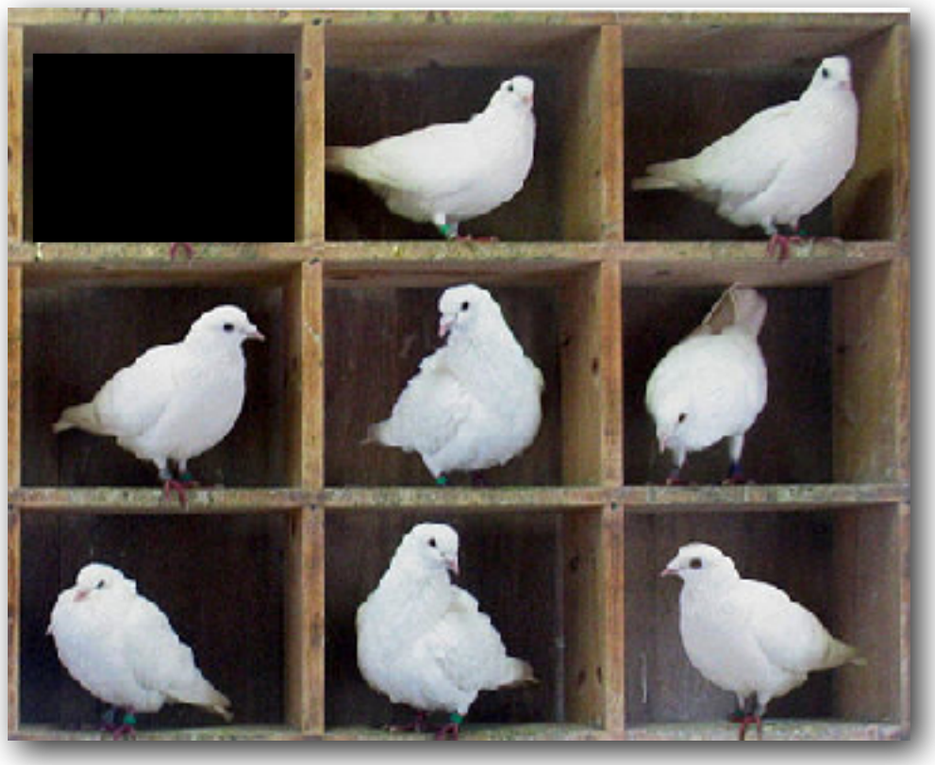
5 partitions
4 edits (*X*)



Approximate matching

Pigeonhole principle: *k+1* pigeons, *k* holes.
At least one has >1 pigeon!

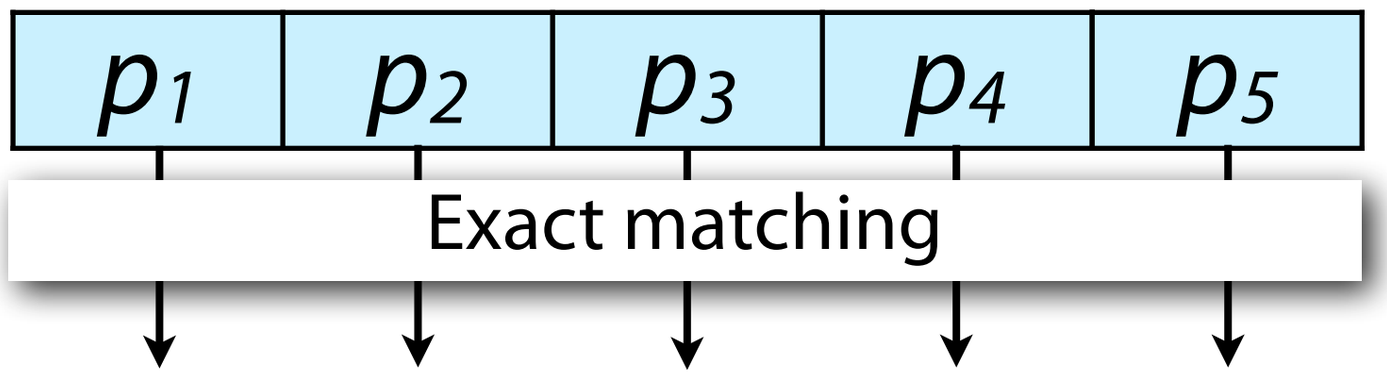
Approximate matching



We have *k* pigeons, *k*+1 holes, at least one...

...is *empty*

Pigeonhole principle



T

What algorithm can we use?

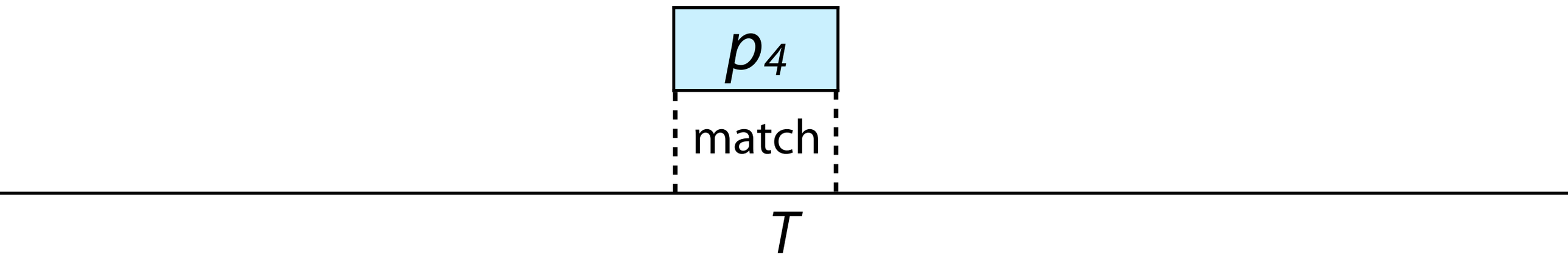
Any exact matching algorithm

If we have a k-mer index, we can use that

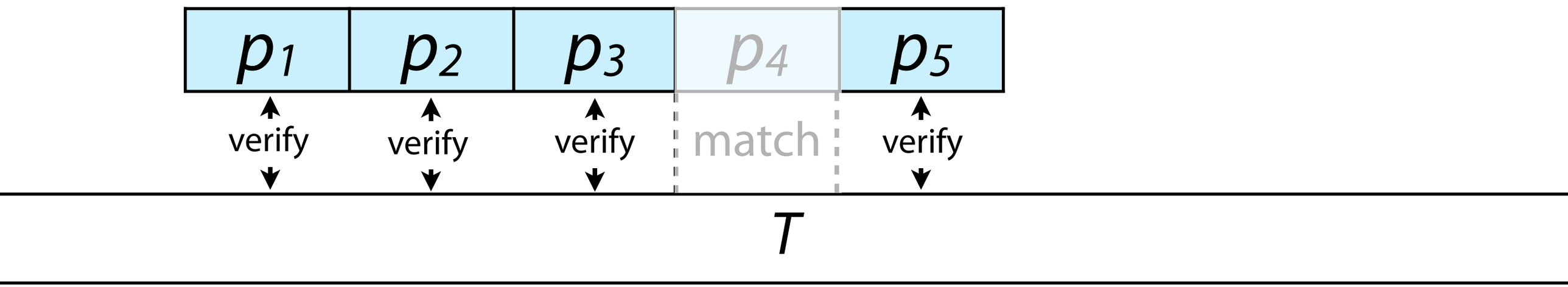
Naive exact matching

Boyer-Moore

Pigeonhole principle

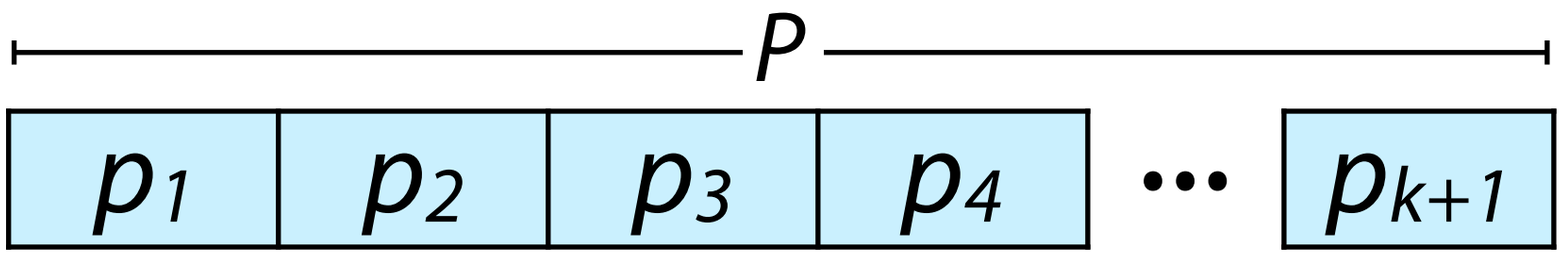


Pigeonhole principle



For Hamming distance, verification is essentially just the inner loop of **naive_approx_hamming** from before

Pigeonhole principle



Advantages

Reuse favorite exact matching algos; fast and easy

Flexible; works for Hamming and edit distance*

Disadvantages

Large *k* yields small partitions matching many times by chance; lots of verification work

k+1 exact matching problems, one per partition

* we don’t know how to do edit distance verification yet

Dynamic Programming & Edit Distance

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Approximate matching for biosequences

Some widely used approximate matching algorithms for DNA (& other strings) came from the biological community, aiming to bring alignments into closer harmony with biological processes that mutate strings

```
Score =  248 bits (129), Expect = 1e-63
Identities = 213/263 (80%), Gaps = 34/263 (12%)
Strand = Plus / Plus

Query: 161 atatcaccacgtcaaaggtgactccaactcca---ccactccattttgttcagataatgc 217
           ||||||||||||||||||||||||||||||  |     | |   || ||||||||||||||
Sbjct: 481 atatcaccacgtcaaaggtgactccaact-tattgatagtgttttatgttcagataatgc 539

Query: 218 ccgatgatcatgtcatgcagctccaccgattgtgagaacgacagcgacttccgtcccagc 277
           |||||||   |||||||||||||||||||| || |            ||||||||||||
Sbjct: 540 ccgatgactttgtcatgcagctccaccgattttg-g-----------ttccgtcccagc 586

Query: 278 c-gtgcc--aggtgctgcctcagattcaggttatgccgctcaattcgctgcgtatatcgc 334
           |  || |  | ||||||||||||||||||||||||||||||||||||| |||||||||
Sbjct: 587 caatgacgta-gtgctgcctcagattcaggttatgccgctcaattcgctgggtatatcgc 645

Query: 335 ttgctgattacgtgcagctttcccttcaggcggga-----------ccagccatccgtc 382
           ||||||||||||||||||||||||||||||||            ||||||||||||
Sbjct: 646 ttgctgattacgtgcagctttcccttcaggcgggattcatacagcggccagccatccgtc 705

Query: 383 ctccatatc-accacgtcaaagg 404
            |||||||| ||||||||||||
Sbjct: 706 atccatatcaaccacgtcaaagg 728
```

Hamming / edit distance

For X, Y where $|X| = |Y|$, *hamming distance* = minimum # substitutions needed to turn one into the other

For X, Y, *edit distance* = minimum # edits (substitutions, insertions, deletions) needed to turn one into the other

Finding distances

```
def hammingDistance(x, y):
    ????
```

Strategy: walk along both strings. For each position, compare the characters in both strings at that position. If not equal, increment hamming distance:

```
x: G C G T A T G C G G C T A A A C G C
   | |   | | | | | | | | |   | | | |
y: G C T T A T G C G G C T A T A C G C
```

Finding distances

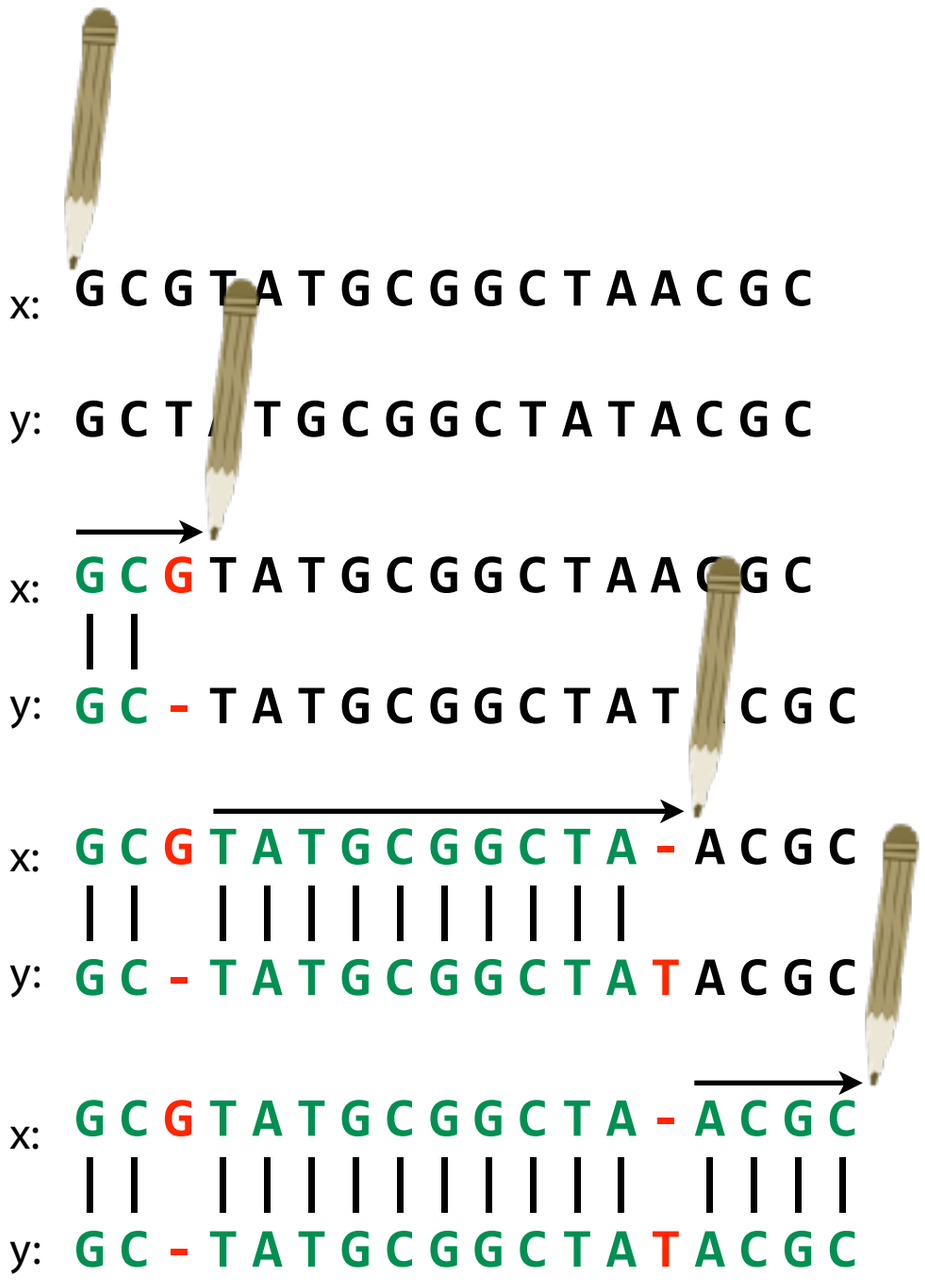
```
def hammingDistance(x, y):
    nmm = 0
    for i in range(0, len(x)):
        if x[i] != y[i]:
            nmm += 1
    return nmm
```

```
def editDistance(x, y):
    ????
```

Finding Hamming distance between strings is pretty easy.
What about edit distance?

Edit distance

Imagine edits are introduced by an *optimal editor* working left-to-right:



Edit transcript summarizes how editor turns *x* into *y*

Operations:
M = match, **R** = replace,
I = insert into *x*, **D** = delete from *x*

MMD

Reminder: this is **D**, not **I**, because we have to delete a character from *x* to make it more like *y*

MMDMMMMMMMMMMI

MMDMMMMMMMMMMIMMMM

Edit distance

Alignments:

```
x: G C G T A T G C G G C T A - A C G C
   |  | | | | | | | | | |   | | | |
y: G C - T A T G C G G C T A T A C G C
```

Edit transcripts (w/r/t *x*):

MMDMMMMMMMMMMIMMMM

Distance = 2

replacement (AKA substitution/mismatch)

```
x: G C G T A T G A G G C T A - A C G C
   |  | | | |   | | | | | |   | | | |
y: G C - T A T G C G G C T A T A C G C
```

MMDMMMMRMMMMMIMMMM

Distance = 3

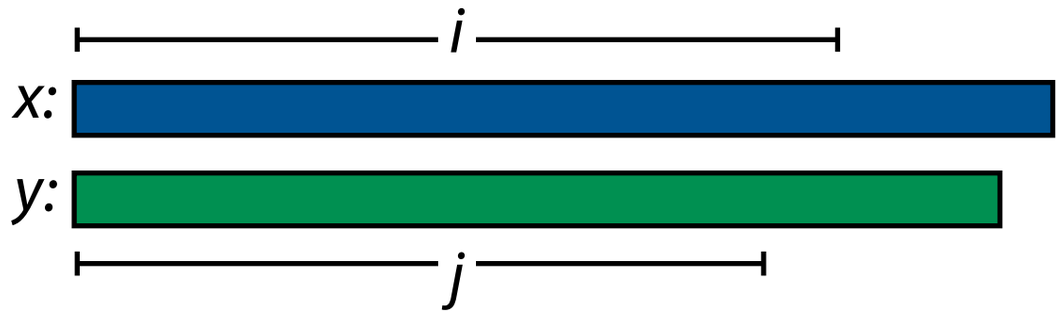
```
x: t h e   l o n g e s t - - - -
         | | | | | | |
y: - - - - l o n g e s t   d a y
```

DDDDMMMMMMMIIII

Distance = 8

Edit distance

D[*i*, *j*]: edit distance between length-*i* prefix of *x* and length-*j* prefix of *y*



Optimal edit transcript for D[*i*, *j*] is built by extending a shorter one by 1 operation. 3 options:

Append **D** to transcript for D[*i*-1, *j*]

Append **I** to transcript for D[*i*, *j*-1]

Append **M** or **R** to transcript for D[*i*-1, *j*-1]

D[*i*, *j*] and its transcript come from 1 of these 3 choices, whichever has fewest edits

D[|*x*|, |*y*|] is the overall edit distance

Edit distance

Is at beginning

Ds at beginning

Let $D[0, j] = j$, and let $D[i, 0] = i$

vertical (D)

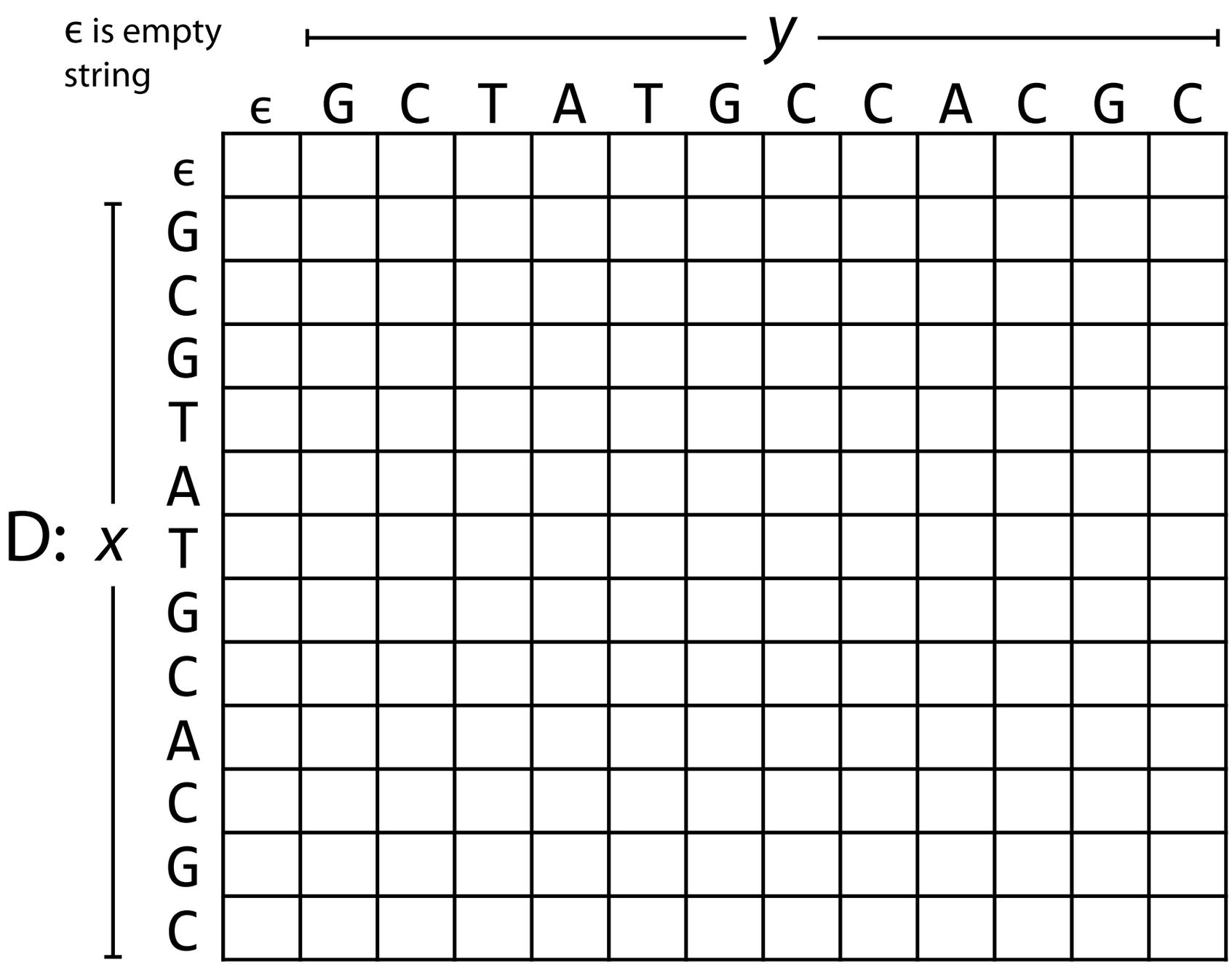
horizontal (I)

diagonal (M or R)

Otherwise, let $D[i, j] = \min \begin{cases} D[i-1, j] + 1 \\ D[i, j-1] + 1 \\ D[i-1, j-1] + \delta(x[i-1], y[j-1]) \end{cases}$

$\delta(a, b)$ is 0 if $a = b$, 1 otherwise

Edit distance: dynamic programming

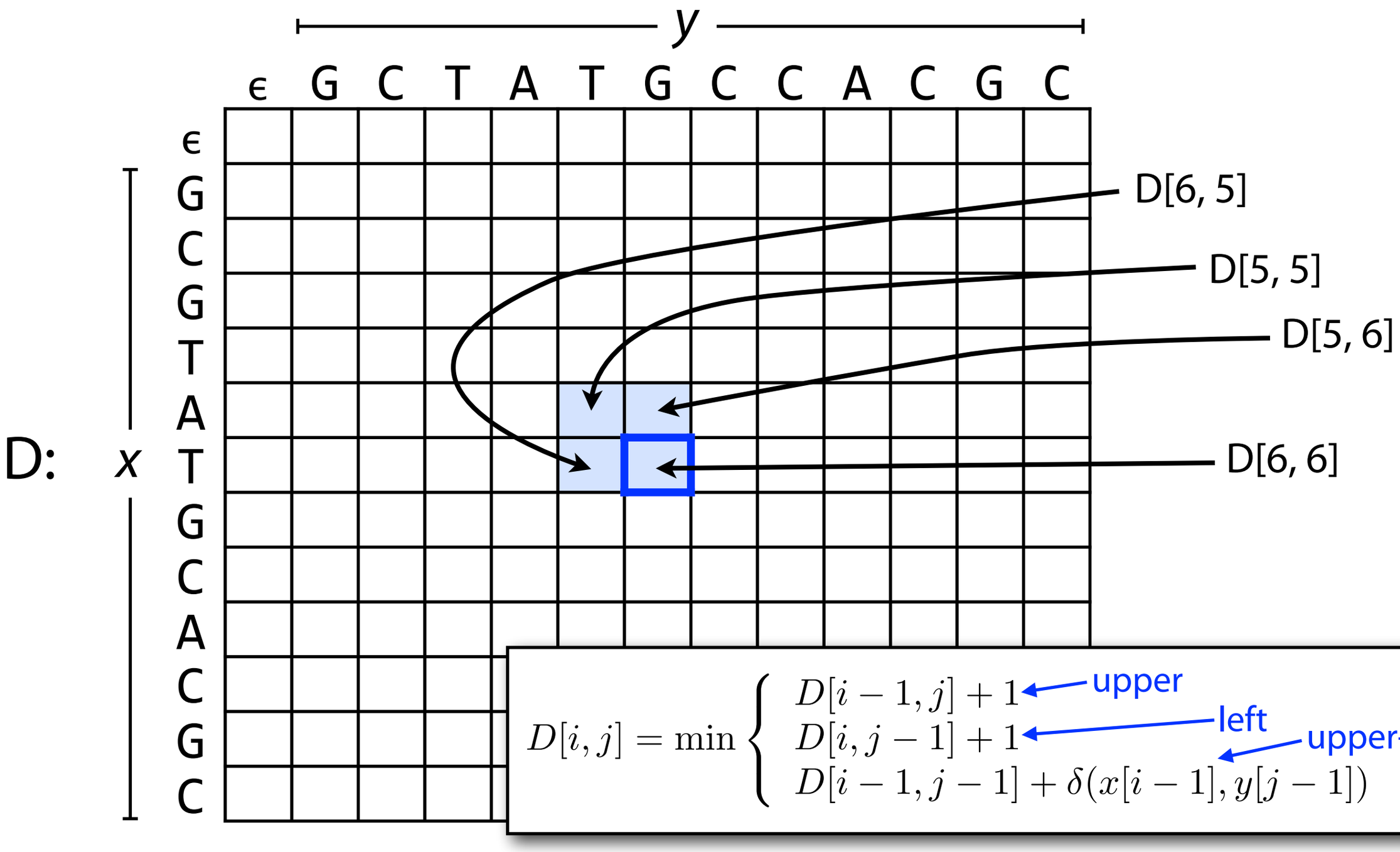


Let $n = |x|$, $m = |y|$

D: $(n+1)$ x $(m+1)$ matrix

D[i, j] = edit distance b/t length-i prefix of x and length-j prefix of y

Edit distance: dynamic programming



Cell depends upon its upper, left, and upper-left neighbors

Edit distance: dynamic programming

First few lines of **edDistDp**:

```
D = numpy.zeros((len(x)+1, len(y)+1), dtype=int)
D[0, 1:] = range(1, len(y)+1)
D[1:, 0] = range(1, len(x)+1)
```

		ϵ	G	C	T	A	T	G	C	C	A	C	G	C
		0	1	2	3	4	5	6	7	8	9	10	11	12
ϵ	0													
G	1													
C	2													
G	3													
T	4													
A	5													
T	6													
G	7													
C	8													
A	9													
C	10													
G	11													
C	12													

Initialize D[0, *j*] to *j*, D[*i*, 0] to *i*

Edit distance: dynamic programming

Loop from **edDistDp**:

```
for i in range(1, len(x)+1):
    for j in range(1, len(y)+1):
        delt = 1 if x[i-1] != y[j-1] else 0
        D[i, j] = min(D[i-1, j-1]+delt, D[i-1, j]+1, D[i, j-1]+1)
```

	ϵ	G	C	T	A	T	G	C	C	A	C	G	C
ϵ	0	1	2	3	4	5	6	7	8	9	10	11	12
G	1												
C	2												
G	3												
T	4												
A	5						etc						
T	6												
G	7												
C	8												
A	9												
C	10												
G	11												
C	12												

Fill remaining cells from top row to bottom and from left to right

Edit distance: dynamic programming

Loop from `edDistDp`:

```
for i in range(1, len(x)+1):
    for j in range(1, len(y)+1):
        delt = 1 if x[i-1] != y[j-1] else 0
        D[i, j] = min(D[i-1, j-1]+delt, D[i-1, j]+1, D[i, j-1]+1)
```

	ϵ	G	C	T	A	T	G	C	C	A	C	G	C
ϵ	0	1	2	3	4	5	6	7	8	9	10	11	12
G	1	?											
C	2												
G	3												
T	4												
A	5												
T	6												
G	7												
C	8												
A	9												
C	10												
G	11												
C	12												

Fill remaining cells from top row to bottom and from left to right

What goes here in `i=1,j=1`?

`x[i-1]` = `y[j-1]` = ‘`G`’,

SO `delt = 0`

```
D[i, j] = min(D[i-1, j-1]+delt,
              D[i-1, j]+1,
              D[i, j-1]+1)
        = min(0 + 0, 1 + 1, 1 + 1)
        = 0
```

Edit distance: dynamic programming

Loop from **edDistDp**:

```
for i in range(1, len(x)+1):
    for j in range(1, len(y)+1):
        delt = 1 if x[i-1] != y[j-1] else 0
        D[i, j] = min(D[i-1, j-1]+delt, D[i-1, j]+1, D[i, j-1]+1)
```

	ϵ	G	C	T	A	T	G	C	C	A	C	G	C
ϵ	0	1	2	3	4	5	6	7	8	9	10	11	12
G	1	0	1	2	3	4	5	6	7	8	9	10	11
C	2	1	0	1	2	3	4	5	6	7	8	9	10
G	3	2	1	1	2	3	3	4	5	6	7	8	9
T	4	3	2	1	2	2	3	4	5	6	7	8	9
A	5	4	3	2	1	2	3	4	5	5	6	7	8
T	6	5	4	3	2	1	2	3	4	5	6	7	8
G	7	6	5	4	3	2	1	2	3	4	5	6	7
C	8	7	6	5	4	3	2	1	2	3	4	5	6
A	9	8	7	6	5	4	3	2	2	2	3	4	5
C	10	9	8	7	6	5	4	3	2	3	2	3	4
G	11	10	9	8	7	6	5	4	3	3	3	2	3
C	12	11	10	9	8	7	6	5	4	4	3	3	2

Fill remaining cells from top row to bottom and from left to right

← Edit distance for x, y

Edit distance: dynamic programming

	ϵ	G	C	T	A	T	G	C	C	A	C	G	C
ϵ	0	1	2	3	4	5	6	7	8	9	10	11	12
G	1	0	1	2	3	4	5	6	7	8	9	10	11
C	2	1	0	1	2	3	4	5	6	7	8	9	10
G	3	2	1	1	2	3	3	4	5	6	7	8	9
T	4	3	2	1	2	2	3	4	5	6	7	8	9
A	5	4	3	2	1	2	3	4	5	5	6	7	8
T	6	5	4	3	2	1	2	3	4	5	6	7	8
G	7	6	5	4	3	2	1	2	3	4	5	6	7
C	8	7	6	5	4	3	2	1	2	3	4	5	6
A	9	8	7	6	5	4	3	2	2	2	3	4	5
C	10	9	8	7	6	5	4	3	2	3	2	3	4
G	11	10	9	8	7	6	5	4	3	3	3	2	3
C	12	11	10	9	8	7	6	5	4	4	3	3	2

← Edit distance for x, y

But where and what are the 2 edits?

Edit distance: getting the alignment

Traceback corresponds to an optimal alignment / edit transcript

At each step, ask: which neighbor (↖, ⇐ or ⇑) gave the minimum?

	ϵ	G	C	T	A	T	G	C	C	A	C	G	C
ϵ	0	1	2	3	4	5	6	7	8	9	10	11	12
G	1	0	1	2	3	4	5	6	7	8	9	10	11
C	2	1	0	1	2	3	4	5	6	7	8	9	10
G	3	2	1	1	2	3	3	4	5	6	7	8	9
T	4	3	2	1	2	2	3	4	5	6	7	8	9
A	5	4	3	2	1	2	3	4	5	5	6	7	8
T	6	5	4	3	2	1	2	3	4	5	6	7	8
G	7	6	5	4	3	2	1	2	3	4	5	6	7
C	8	7	6	5	4	3	2	1	2	3	4	5	6
A	9	8	7	6	5	4	3	2	2	2	3	4	5
C	10	9	8	7	6	5	4	3	2	3	2	3	4
G	11	10	9	8	7	6	5	4	3	3	3	2	3
C	12	11	10	9	8	7	6	5	4	4	3	3	2

A: From here

Q: How did I get here?

Edit distance: getting the alignment

Traceback corresponds to an optimal alignment / edit transcript

At each step, ask: which neighbor (↖, ⇐ or ⇑) gave the minimum?

	ϵ	G	C	T	A	T	G	C	C	A	C	G	C
ϵ	0	1	2	3	4	5	6	7	8	9	10	11	12
G	1	0	1	2	3	4	5	6	7	8	9	10	11
C	2	1	0	1	2	3	4	5	6	7	8	9	10
G	3	2	1	1	2	3	3	4	5	6	7	8	9
T	4	3	2	1	2	2	3	4	5	6	7	8	9
A	5	4	3	2	1	2	3	4	5	5	6	7	8
T	6	5	4	3	2	1	2	3	4	5	6	7	8
G	7	6	5	4	3	2	1	2	3	4	5	6	7
C	8	7	6	5	4	3	2	1	2	3	4	5	6
A	9	8	7	6	5	4	3	2	2	2	3	4	5
C	10	9	8	7	6	5	4	3	2	3	2	3	4
G	11	10	9	8	7	6	5	4	3	3	3	2	3
C	12	11	10	9	8	7	6	5	4	4	3	3	2

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G	1	0	1	2	3	4	5	6	7	8	9	10	11
C	2	1	0	1	2	3	4	5	6	7	8	9	10
G	3	2	1	1	2	3	3	4	5	6	7	8	9
T	4	3	2	1	2	2	3	4	5	6	7	8	9
A	5	4	3	2	1	2	3	4	5	5	6	7	8
T	6	5	4	3	2	1	2	3	4	5	6	7	8
G	7	6	5	4	3	2	1	2	3	4	5	6	7
C	8	7	6	5	4	3	2	1	2	3	4	5	6
A	9	8	7	6	5	4	3	2	2	2	3	4	5
C	10	9	8	7	6	5	4	3	2	3	2	3	4
G	11	10	9	8	7	6	5	4	3	3	3	2	3
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ϵ	0	1	2	3	4	5	6	7	8	9	10	11	12
G	1	0	1	2	3	4	5	6	7	8	9	10	11
C	2	1	0	1	2	3	4	5	6	7	8	9	10
G	3	2	1	1	2	3	3	4	5	6	7	8	9
T	4	3	2	1	2	2	3	4	5	6	7	8	9
A	5	4	3	2	1	2	3	4	5	5	6	7	8
T	6	5	4	3	2	1	2	3	4	5	6	7	8
G	7	6	5	4	3	2	1	2	3	4	5	6	7
C	8	7	6	5	4	3	2	1	2	3	4	5	6
A	9	8	7	6	5	4	3	2	2	2	3	4	5
C	10	9	8	7	6	5	4	3	2	3	2	3	4
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C	12	11	10	9	8	7	6	5	4	4	3	3	2

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Traceback corresponds to an optimal alignment / edit transcript

At each step, ask: which neighbor (↖, ⇐ or ⇑) gave the minimum?

	ϵ	G	C	T	A	T	G	C	C	A	C	G	C
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C	2	1	0	1	2	3	4	5	6	7	8	9	10
G	3	2	1	1	2	3	3	4	5	6	7	8	9
T	4	3	2	1	2	2	3	4	5	6	7	8	9
A	5	4	3	2	1	2	3	4	5	5	6	7	8
T	6	5	4	3	2	1	2	3	4	5	6	7	8
G	7	6	5	4	3	2	1	2	3	4	5	6	7
C	8	7	6	5	4	3	2	1	2	3	4	5	6
A	9	8	7	6	5	4	3	2	2	2	3	4	5
C	10	9	8	7	6	5	4	3	2	3	3	4	4
G	11	10	9	8	7	6	5	4	3	3	3	3	3
C	12	11	10	9	8	7	6	5	4	4	3	3	2

Alignment:

```
GCGTATG-CACGC
||  |||| |||||
GC-TATGCCACGC
```

Edit transcript:

MMDMMMMIMMMMM

Edit distance: getting the alignment

To find the edit distance and the alignment we did:
(a) table filling, (b) traceback

	ϵ	G	C	T	A	T	G	C	C	A	C	G	C
ϵ	0	1	2	3	4	5	6	7	8	9	10	11	12
G	1	0	1	2	3	4	5	6	7	8	9	10	11
C	2	1	0	1	2	3	4	5	6	7	8	9	10
G	3	2	1	1	2	3	3	4	5	6	7	8	9
T	4	3	2	1	2	2	3	4	5	6	7	8	9
A	5	4	3	2	1	2	3	4	5	5	6	7	8
T	6	5	4	3	2	1	2	3	4	5	6	7	8
G	7	6	5	4	3	2	1	2	3	4	5	6	7
C	8	7	6	5	4	3	2	1	2	3	4	5	6
A	9	8	7	6	5	4	3	2	2	2	3	4	5
C	10	9	8	7	6	5	4	3	2	3	2	3	4
G	11	10	9	8	7	6	5	4	3	3	3	2	3
C	12	11	10	9	8	7	6	5	4	4	3	3	2

If $|X| = m$ and $|Y| = n$, how much work is table filling?

Filling (m + 1)(n + 1) cells, each requiring constant work, so $O(mn)$

How much work is traceback?

Each step goes ↖, ⇐ or ⇑.

Worst case: traceback never moves diagonally, requiring m ⇑'s and n ⇐'s, so $O(m + n)$

Edit distance: summary

Matrix-filling dynamic programming algorithm is $O(mn)$ time and space

- FillIng matrix is $O(mn)$ space and time, yields edit distance
- Traceback is $O(m + n)$ time, yields optimal alignment / edit transcript

Approximate matching with edit distance

Can we search for the occurrence of *P* in *T* with least edits?

		T																				
		ϵ	A	A	C	C	C	T	A	T	G	T	C	A	T	G	C	C	T	T	G	G	A
	ϵ	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
	T	**1**																					
	A	**2**																					
	C	**3**																					
	G	**4**																					
P	T	**5**																					
	C	**6**																					
	A	**7**																					
	G	**8**																					
	C	**9**																					

Approximate matching with edit distance

1. Initialize first row with 0's rather than increasing integers, then fill matrix

			T																		
		ϵ	A	A	C	C	C	T	A	T	G	T	C	A	T	G	C	C	T	T	G	G	A
	ϵ	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	T	1																					
	A	2																					
	C	3																					
	G	4																					
P	T	5																					
	C	6																					
	A	7																					
	G	8																					
	C	9																					

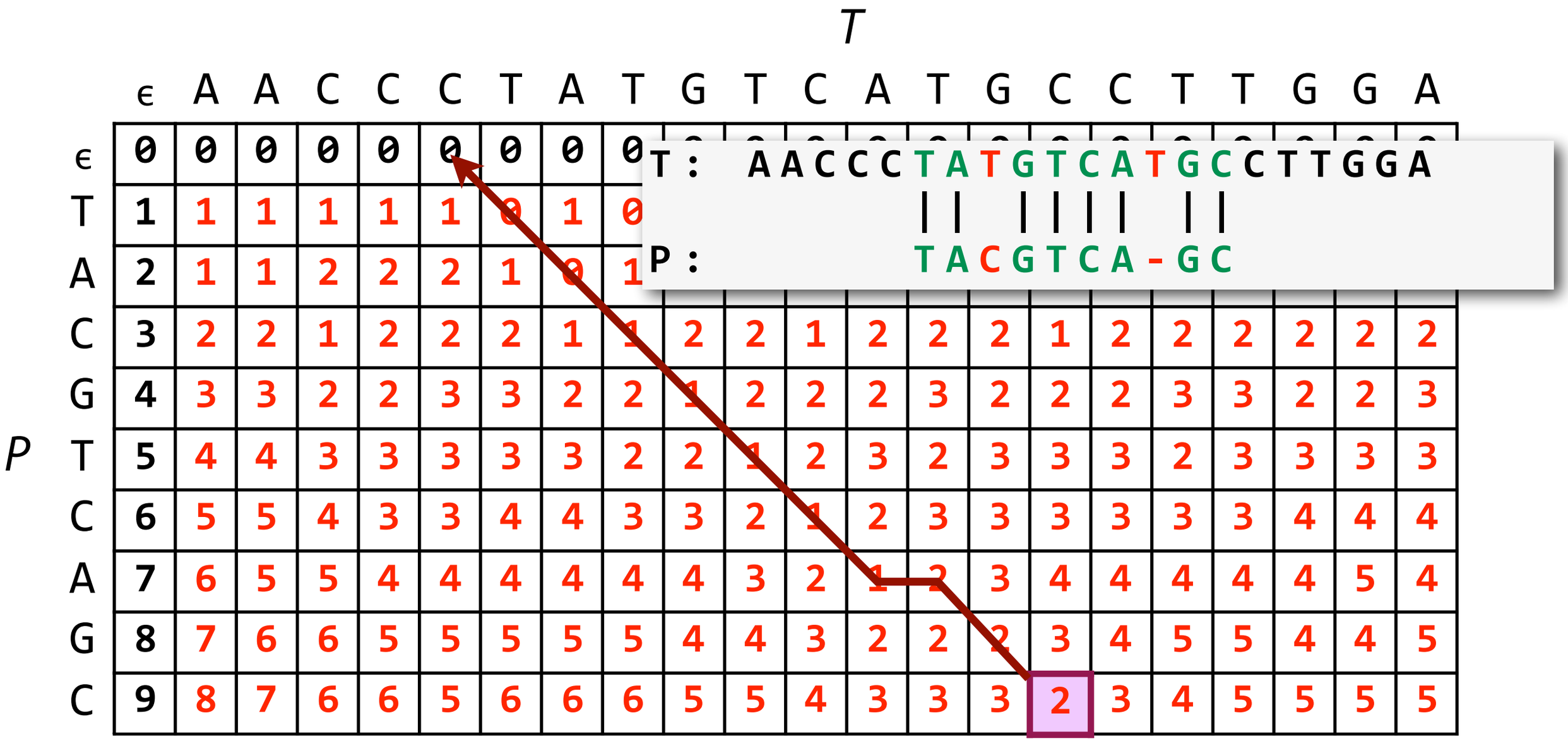
Approximate matching with edit distance

1. Initialize first row with 0's rather than increasing integers, then fill matrix

			T																		
		ϵ	A	A	C	C	C	T	A	T	G	T	C	A	T	G	C	C	T	T	G	G	A
	ϵ	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	T	1	1	1	1	1	1	0	1	0	1	0	1	1	0	1	1	1	0	0	1	1	1
	A	2	1	1	2	2	2	1	0	1	1	1	1	1	1	1	2	2	1	1	1	2	1
	C	3	2	2	1	2	2	2	1	1	2	2	1	2	2	2	1	2	2	2	2	2	2
	G	4	3	3	2	2	3	3	2	2	1	2	2	2	3	2	2	2	3	3	2	2	3
P	T	5	4	4	3	3	3	3	3	2	2	1	2	3	2	3	3	3	2	3	3	3	3
	C	6	5	5	4	3	3	4	4	3	3	2	1	2	3	3	3	3	3	3	4	4	4
	A	7	6	5	5	4	4	4	4	4	4	3	2	1	2	3	4	4	4	4	4	5	4
	G	8	7	6	6	5	5	5	5	5	4	4	3	2	2	2	3	4	5	5	4	4	5
	C	9	8	7	6	6	5	6	6	6	5	5	4	3	3	3	2	3	4	5	5	5	5

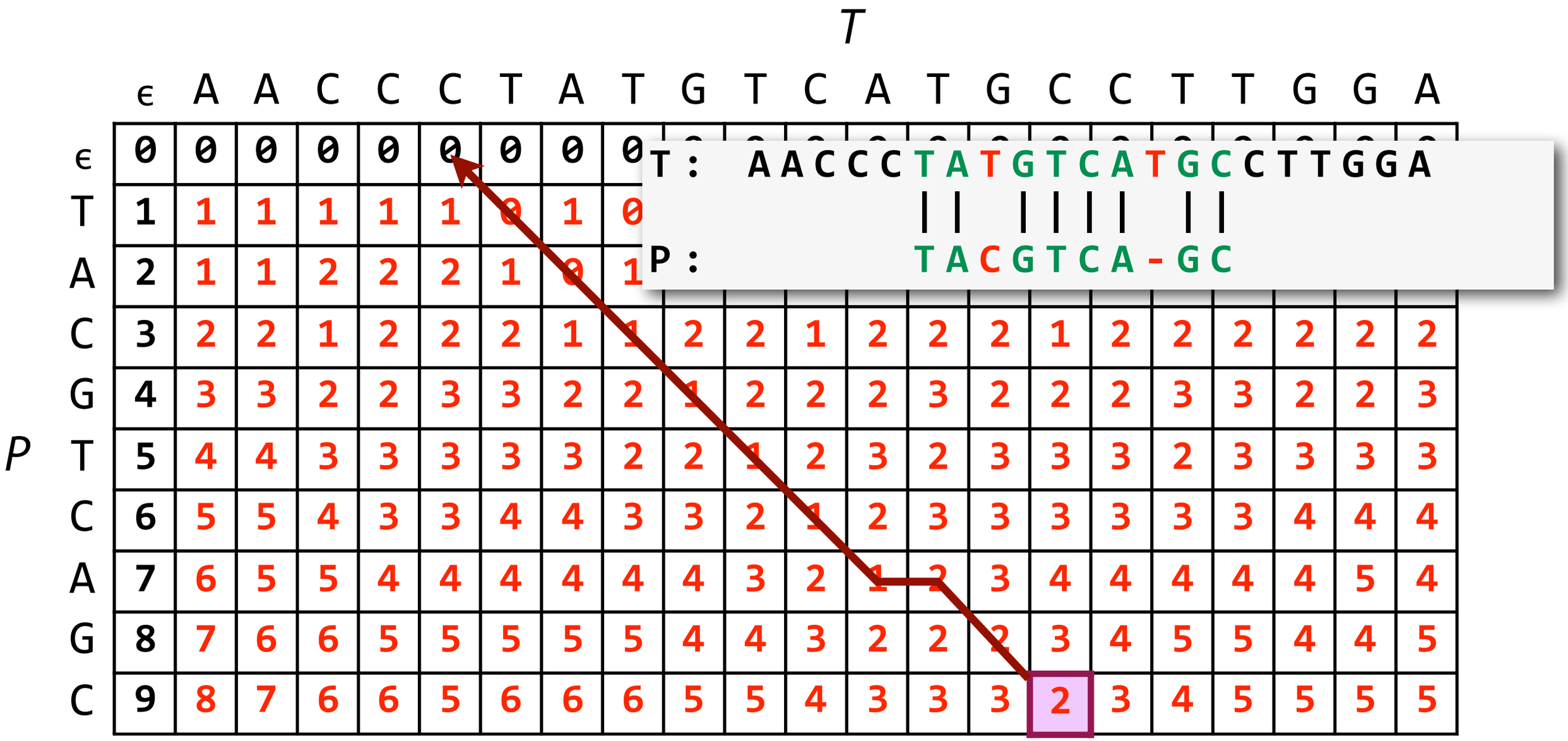
Approximate matching with edit distance

1. Initialize first row with 0's rather than increasing integers, then fill matrix
2. Pick lowest edit distance *in the bottom row*, traceback to top row



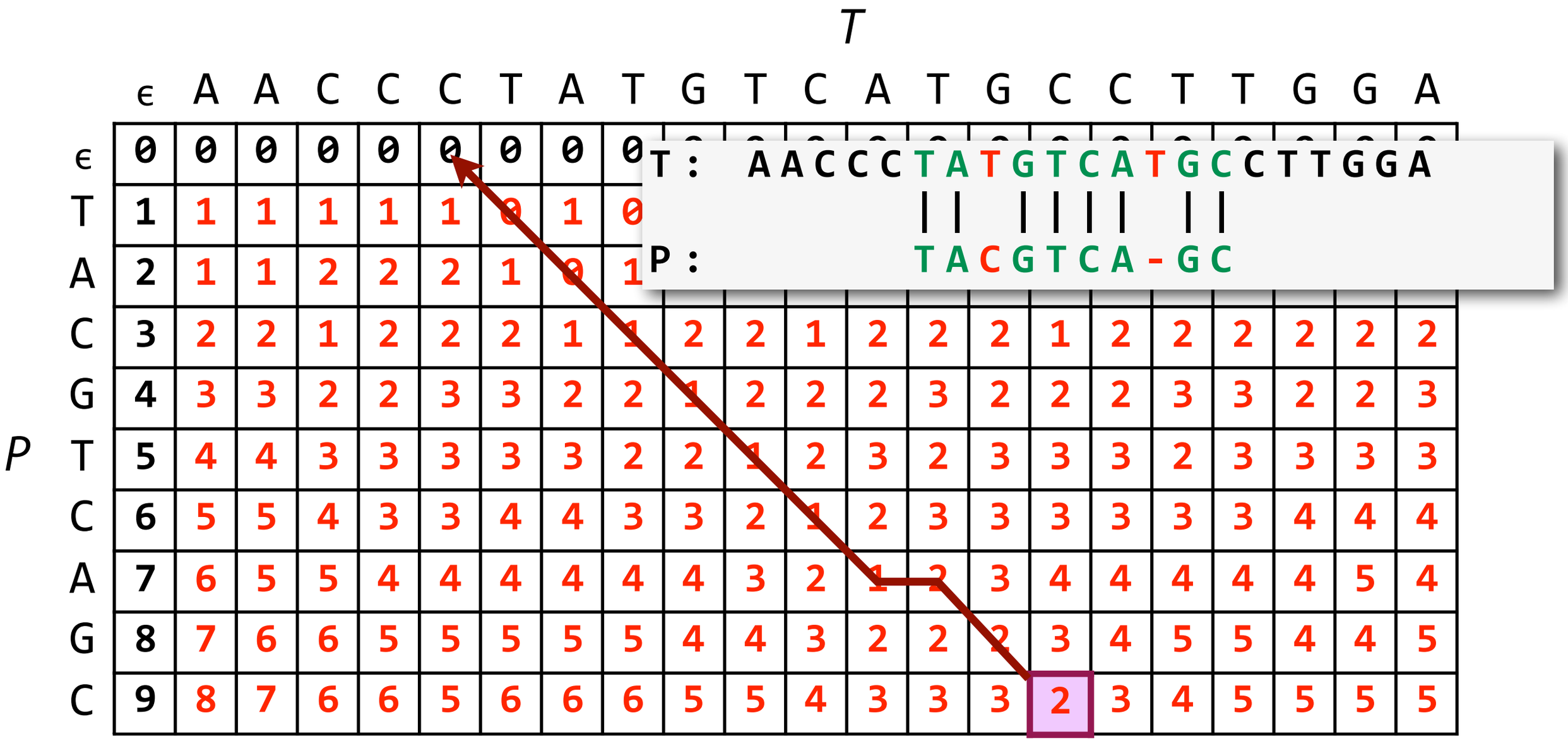
Approximate matching with edit distance

D[*i, j*] equals the optimal edit distance between the length-*i* prefix of *P* and...



Approximate matching with edit distance

D[*i, j*] equals the optimal edit distance between the length-*i* prefix of *P* and a substring of *T* ending at position *j*.



Index-Assisted Approximate Matching

Ben Langmead

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WHITING SCHOOL
of ENGINEERING

Department of Computer Science

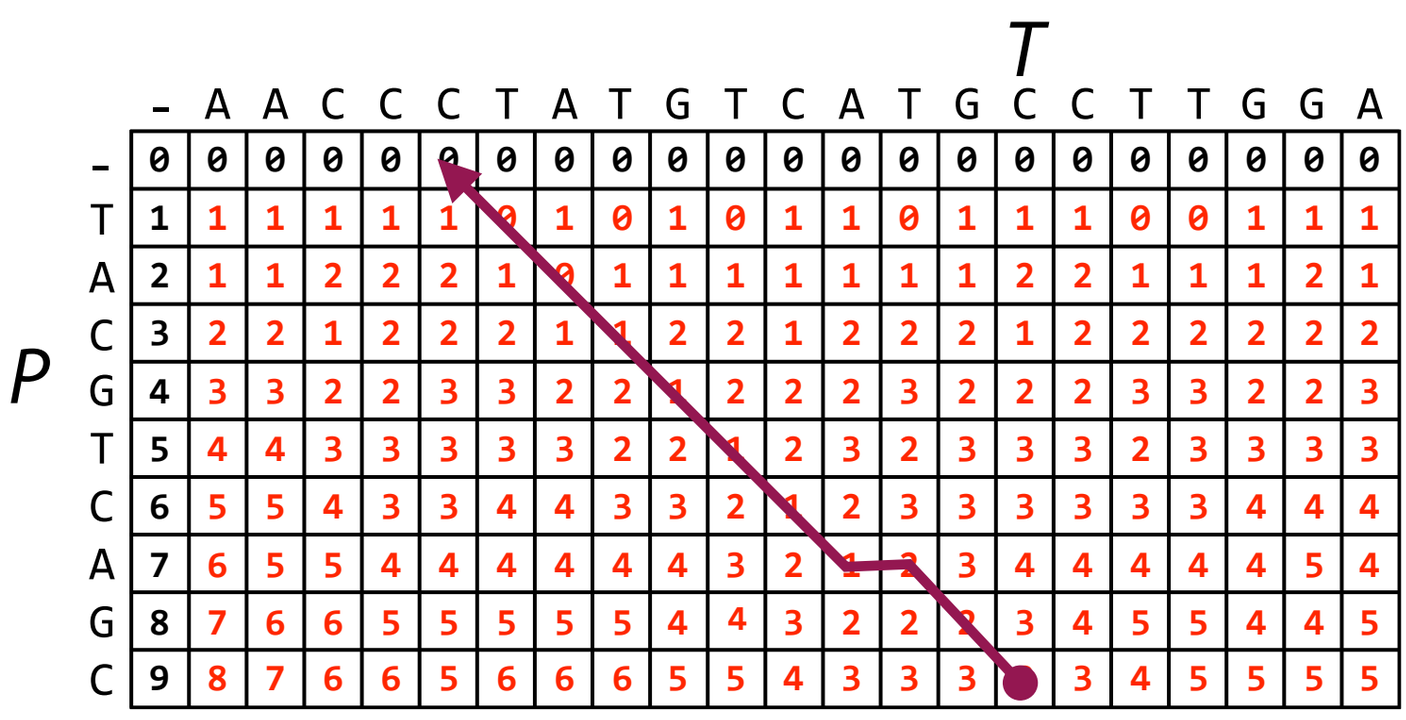
Please sign guestbook (www.langmead-lab.org/teaching-materials) to tell me briefly how you are using the slides. For original Keynote files, email me (ben.langmead@gmail.com).

Dynamic programming summary

A powerful set of tools:

- DP deals naturally with both mismatches and gaps
- DP scoring can take into account variation, sequencing error, etc

Along the way we saw an algorithm we might use for read alignment:



Finding approximate occurrences of *P* in *T*

...but no faster than $O(mn)$ and m is big!!!

A de-motivating example

d = 6 x 10^9 reads
n = 100 nt
m = 3 x 10^9 nt ≈ human

} ≈ 1 week-long run of

Illumina HiSeq 2000

Say we have 1,000 processors, each clocked at 3 GHz, each capable of completing 8 dynamic programming cell updates per clock cycle

(We're being optimistic)

Total of **d** x **m** x **n** = 2 x 10^{21} cell updates

Takes > 2 years

A de-motivating example

d = 6 x 10^9 reads
n = 100 nt
m = 3 x 10^9 nt

} ≈ 1 week-long run of human

Illumina HiSeq 2000

Problem: our dynamic programming approach is $O(dmn)$

We'll now consider two ideas for how to maintain the power of dynamic programming while diminishing effect of *m*

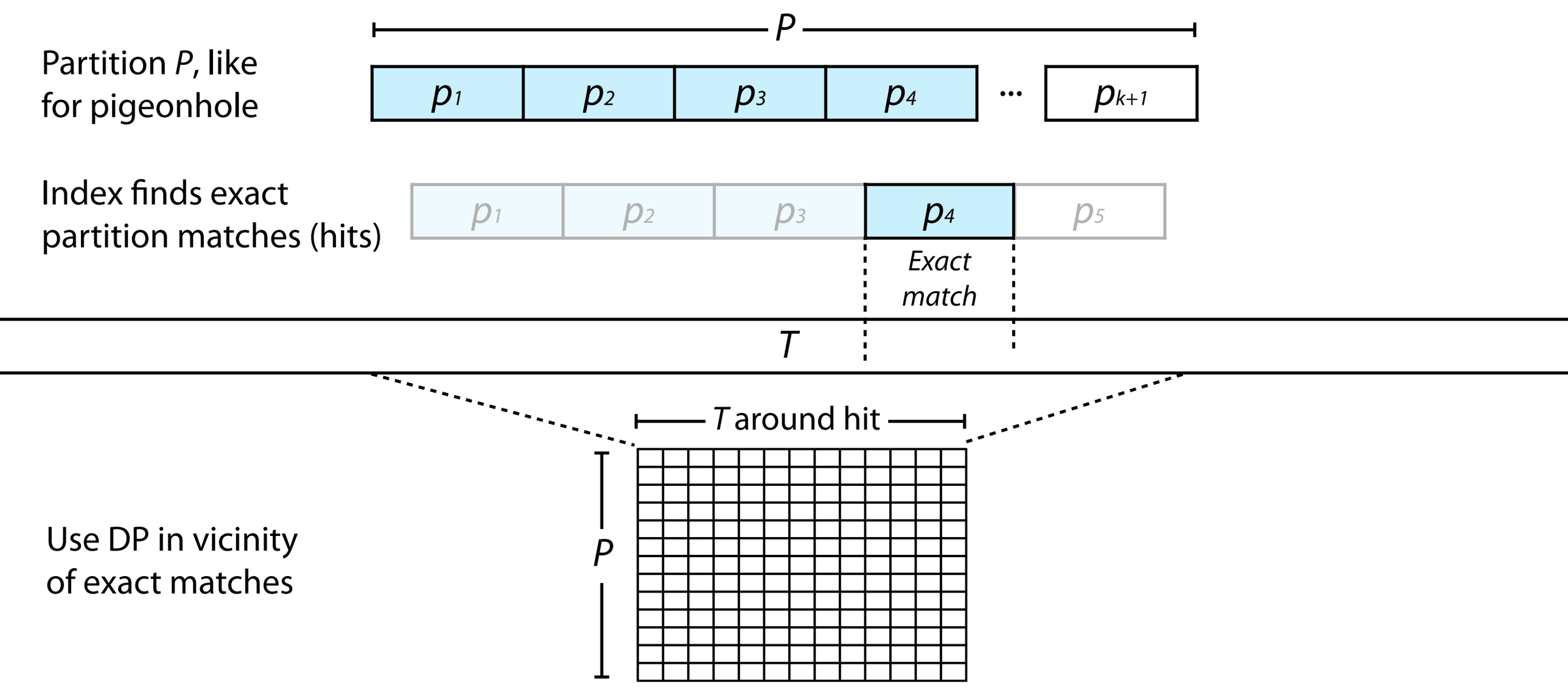
Index-assisted approximate matching

Idea 1: Use index for exact-matching subproblems, follow up with DP

Partition *P*, like for pigeonhole

Index finds exact partition matches (hits)

Use DP in vicinity of exact matches



Index-assisted approximate matching

Index-assisted function for finding occurrences of *P* in *T* with up to *k* edits:

```
def queryIndexEdit(p, t, k, index):
    ''' Look for occurrences of p in t with up to k edits using an
        index combined with dynamic-programming alignment. '''
    l = index.ln
    occurrences = []
    seen = set()     # for avoiding reporting same hit twice
    for part, poff in partition(p, k+1):
        for hit in index.occurrences(part): # query index w/ partition
            # left edge of T to include in DP matrix
            lf = max(0, hit - poff - k)
            # right edge of T to include in DP matrix
            rt = min(len(t), hit - poff + len(p) + k)
            mn, off, xcript = kEditDp(p, t[lf:rt])
            off += lf
            if mn <= k and (mn, off) not in seen:
                occurrences.append((mn, off, xcript))
                seen.add((mn, off))
return occurrences
```

Partition P

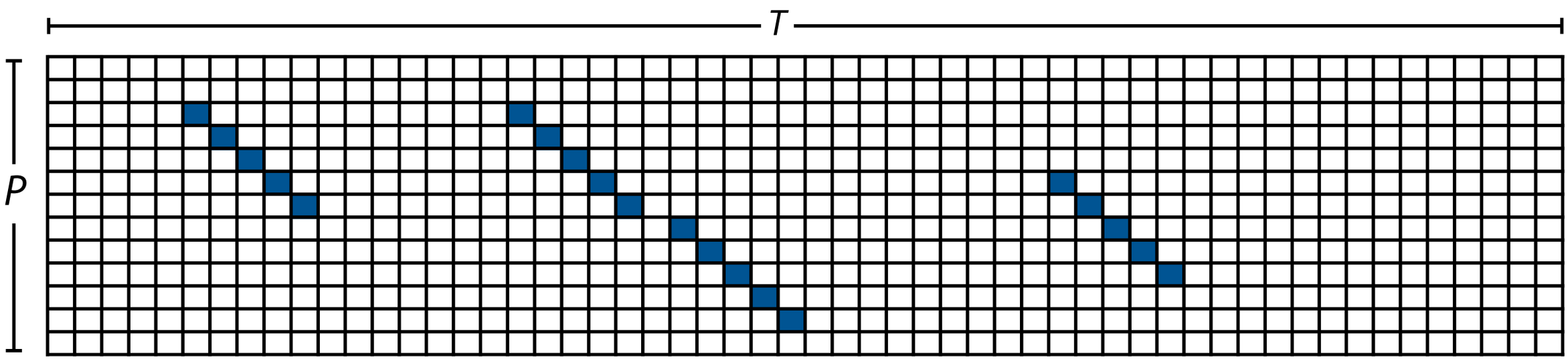
Query index

Dynamic programming

Python example: http://bit.ly/CG_kEdit_idx

Index-assisted approximate matching

Think in terms of the full *P*-to-*T* dynamic programming matrix



Index is identifying diagonal stretches of matches

These are likely to be part of a high-scoring alignment

Many stretches within a few diagonals of each other are even more likely to be part of a high-scoring alignment

Neighborhood search

Idea 2: Use index to find exact occurrences of strings in *P*'s *neighborhood*

Neighborhood = set of strings within some Hamming / edit distance

The 1-edit neighborhood of cat, using DNA alphabet:

cat, aat, gat, tat, cct, cgt, ctt, caa, cac, cag, ca, ct, at, acat, ccat, gcat, tcat, ...

All ways to add 1 mismatch — All ways to delete 1 char — All ways to insert 1 char — ...

The 2-mismatch neighborhood of cat:

cat, aat, gat, tat, cct, cgt, ctt, caa, cac, cag

- All ways to add 1 mismatch
  - All ways to add 2nd mismatch to aat
  - All ways to add 2nd mismatch to gat
  - ...

Neighborhood search

Idea 2: Use index to find occurrences of strings in *P*'s "neighborhood"

Is the neighborhood huge? Can we bound it?

If $|P| = n$, and $|\Sigma| = a$, how many strings are within Hamming distance 1?

$$1 + \underbrace{n(a - 1)}$$

P itself

a - 1 ways to replace each of *P*'s *n* chars

How many strings are within edit distance 1?

$$1 + n(a - 1) + n + \underbrace{(n + 1)a}$$

Delete each char in *P* *

n + 1 positions where we can insert any of the *a* characters *

In both cases, $O(an)$ strings in the neighborhood

* Some insertions are equivalent. E.g. there are two equivalent insertions of 'a' into 'cat'. Likewise deletions ('caat').

Neighborhood search

How about within Hamming or edit distance 2?

$O(an)$ strings within Hamming or edit distance 1, each with $O(an)$ neighbors within distance 1, so $O(a^2n^2)$

Within distance k?

$O(a^kn^k)$

How much work to query suffix tree with all strings within distance k?

$O(n$ + # occurrences) for each of the $O(a^kn^k)$ strings, so roughly $O(a^kn^{k+1})$

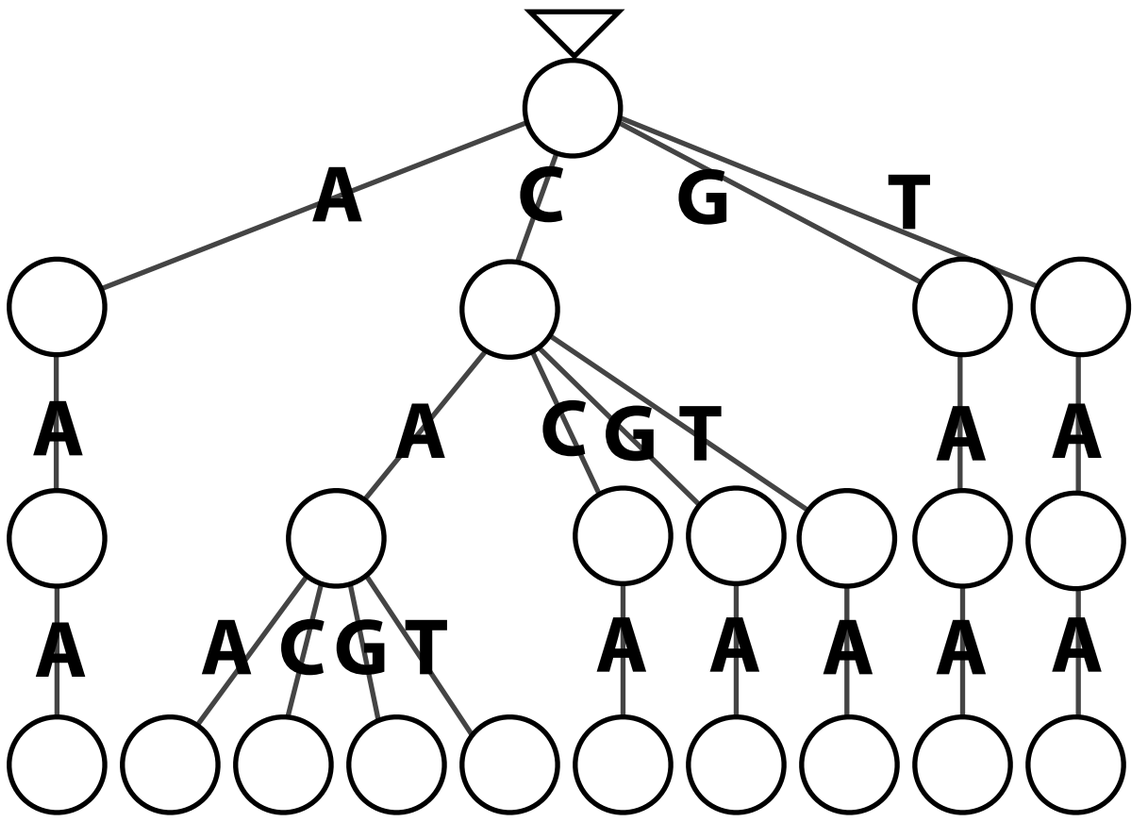
Compare to $O(a^kn^{k+1})$ to $O(mn)$ for full dynamic programming

Good: no m Bad: exponential in k

Neighborhood search

Organize neighborhood of *P* into a trie

Neighbors of P = CAA, within hamming distance 1:

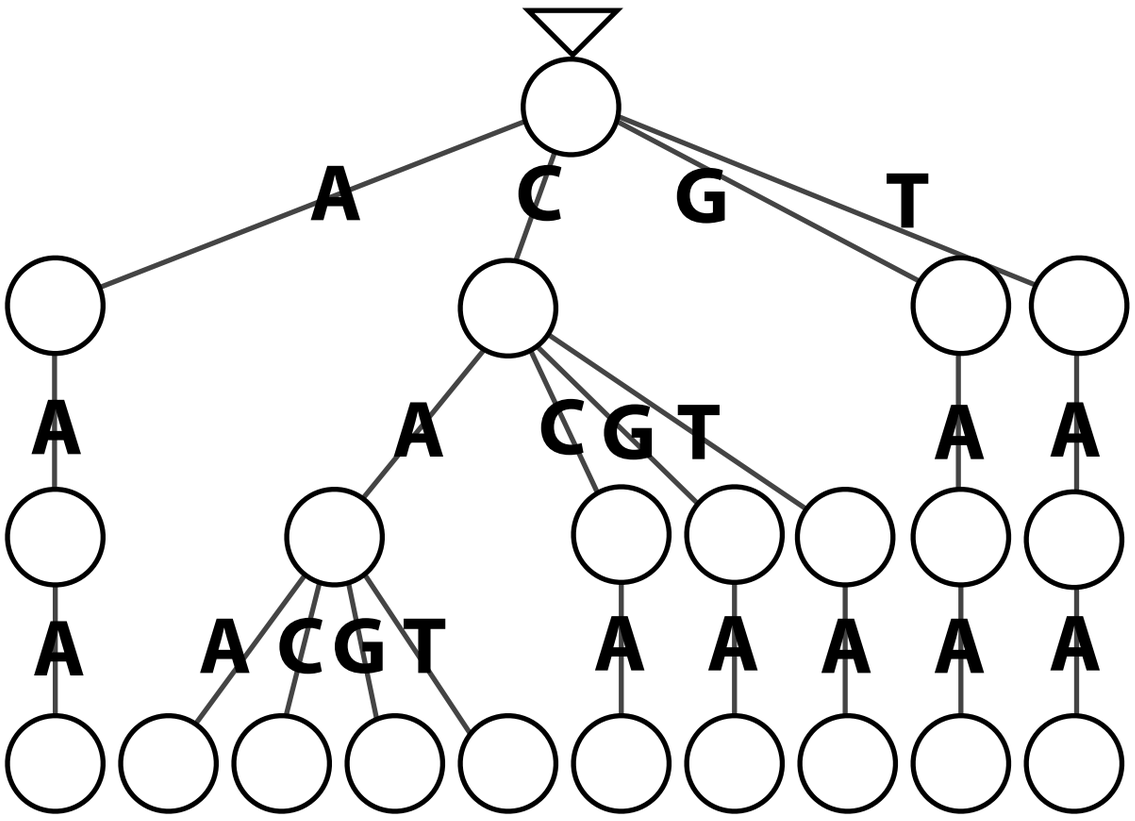


leaves = # neighbors =
$1 + n(a - 1) = 1 + 3(4 - 1) = 10$

Neighborhood search

Navigating and/or building neighborhood trie is simple with recursion

Assume Hamming distance for now: Move left-to-right across *P* and start with "budget" of *k* mismatches



At each step, for each alphabet character *c*:

- If *c* matches current character in *P*, recursively build subtree starting at next position of *P* with same budget

- If c *mis*matches current character in P *and* budget > 0, recursively build subtree starting at next position of *P* with 1 subtracted from budget. Otherwise if budget = 0, move on.

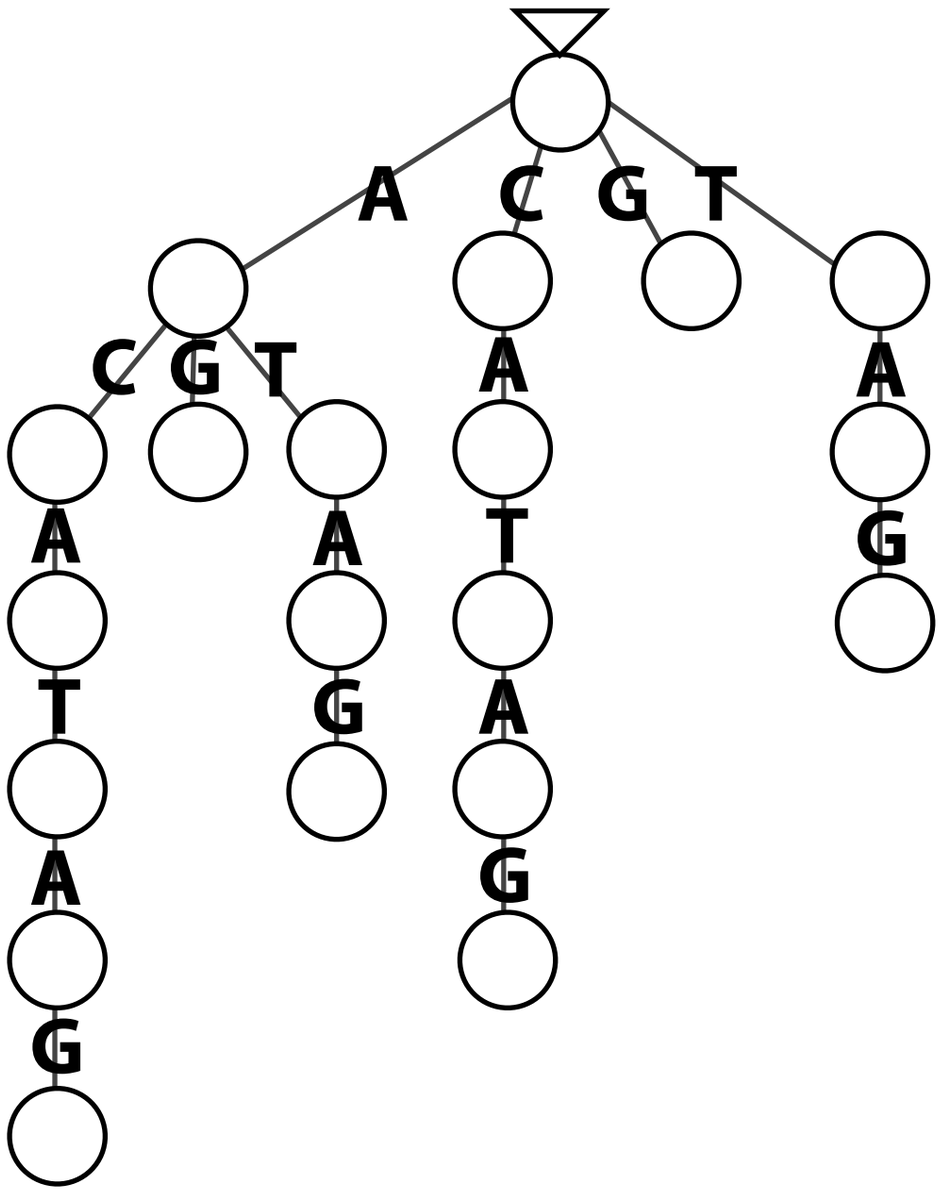
Neighborhood search

```
>>> stringNeighbors("cat", "acgt", edits=1, gaps=False)
['aat', 'gat', 'tat', 'cct', 'cgt', 'ctt', 'caa', 'cac', 'cag', 'cat']
>>> stringNeighbors("cat", "acgt", edits=1, gaps=True)
['acat', 'ccat', 'gcat', 'tcat', 'at', 'aat', 'gat', 'tat', 'caat',
'ccat', 'cgat', 'ctat', 'ct', 'cct', 'cgt', 'ctt', 'caat', 'cact',
'cagt', 'catt', 'ca', 'caa', 'cac', 'cag', 'cata', 'catc', 'catg',
'catt', 'cat']
```

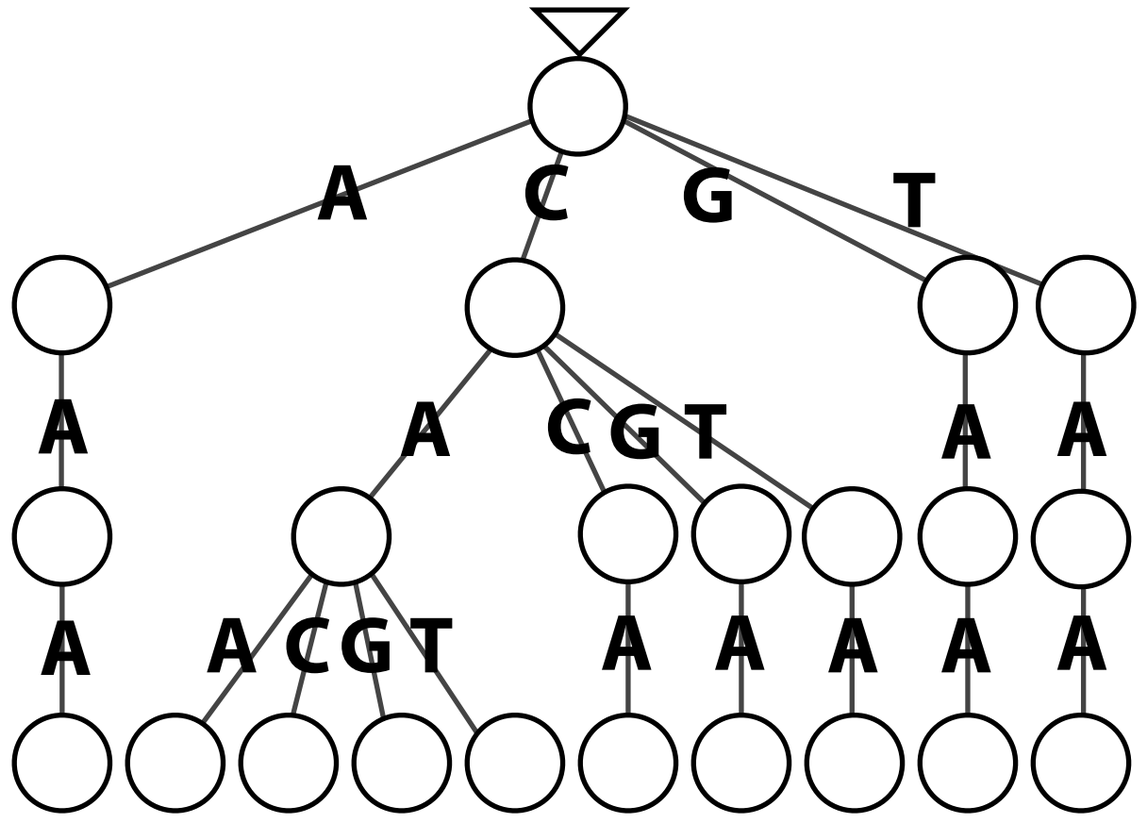
Co-traversal

Say index is a suffix trie. Imagine querying it with each neighbor.

Suffix trie of T = ACATAG



Trie for neighborhood within 1 mismatch of P = CAA



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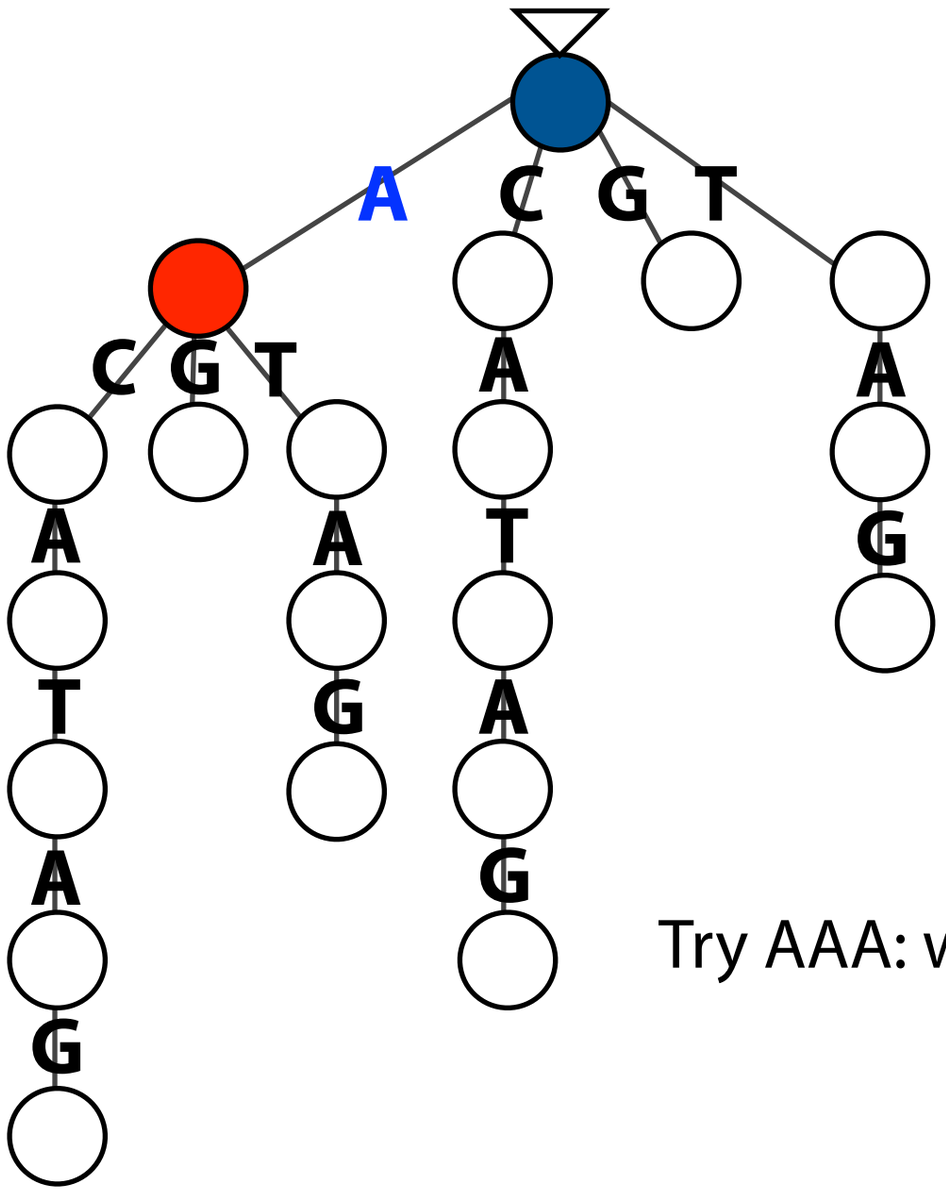
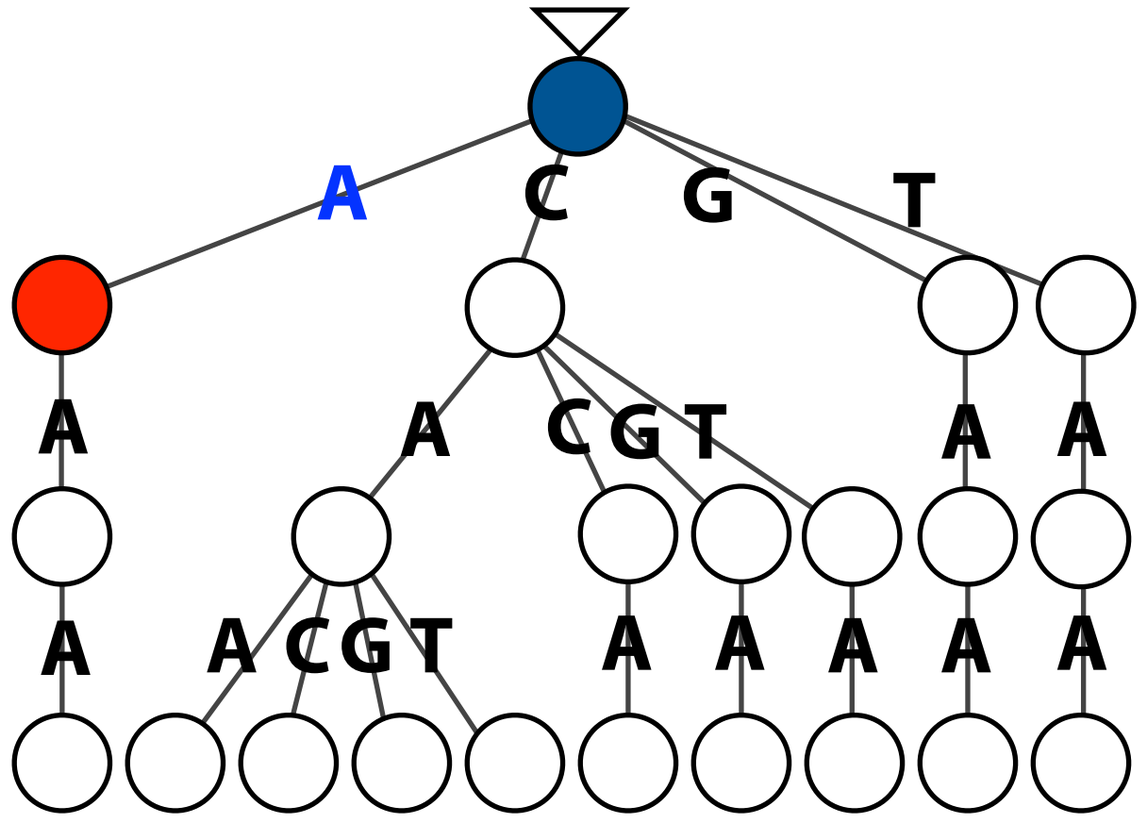


Trie for neighborhood within 1 mismatch of P = CAA

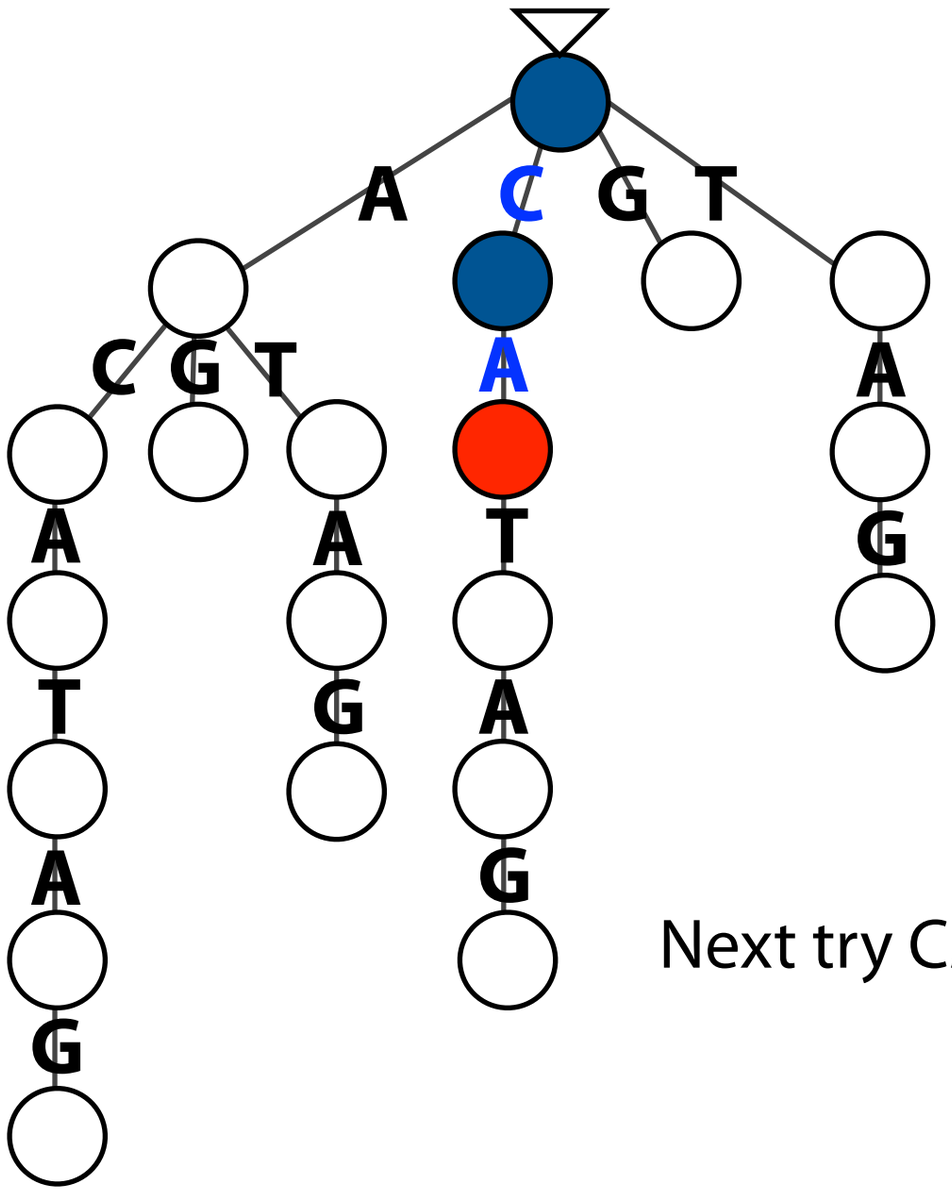


Try AAA: we fall off suffix trie after A, before AA

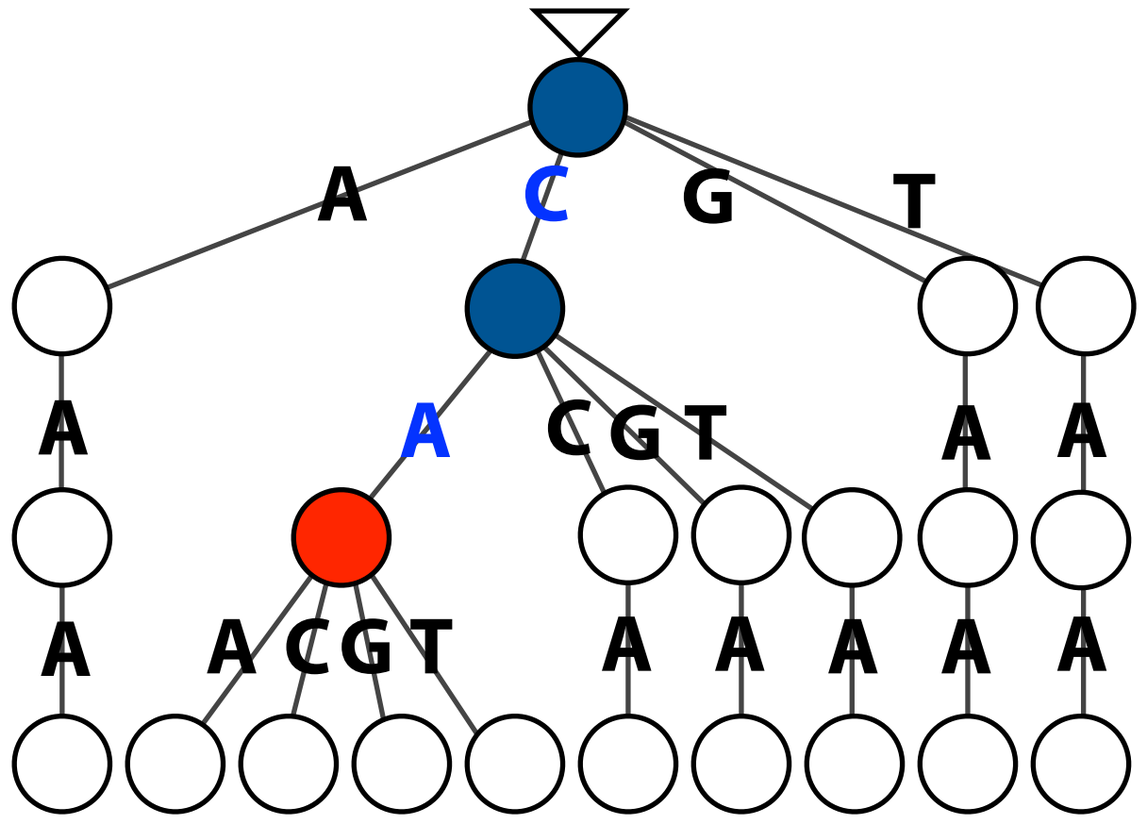
Co-traversal

Say index is a suffix trie. Imagine querying it with each neighbor.

Suffix trie of T = ACATAG



Trie for neighborhood within 1 mismatch of P = CAA

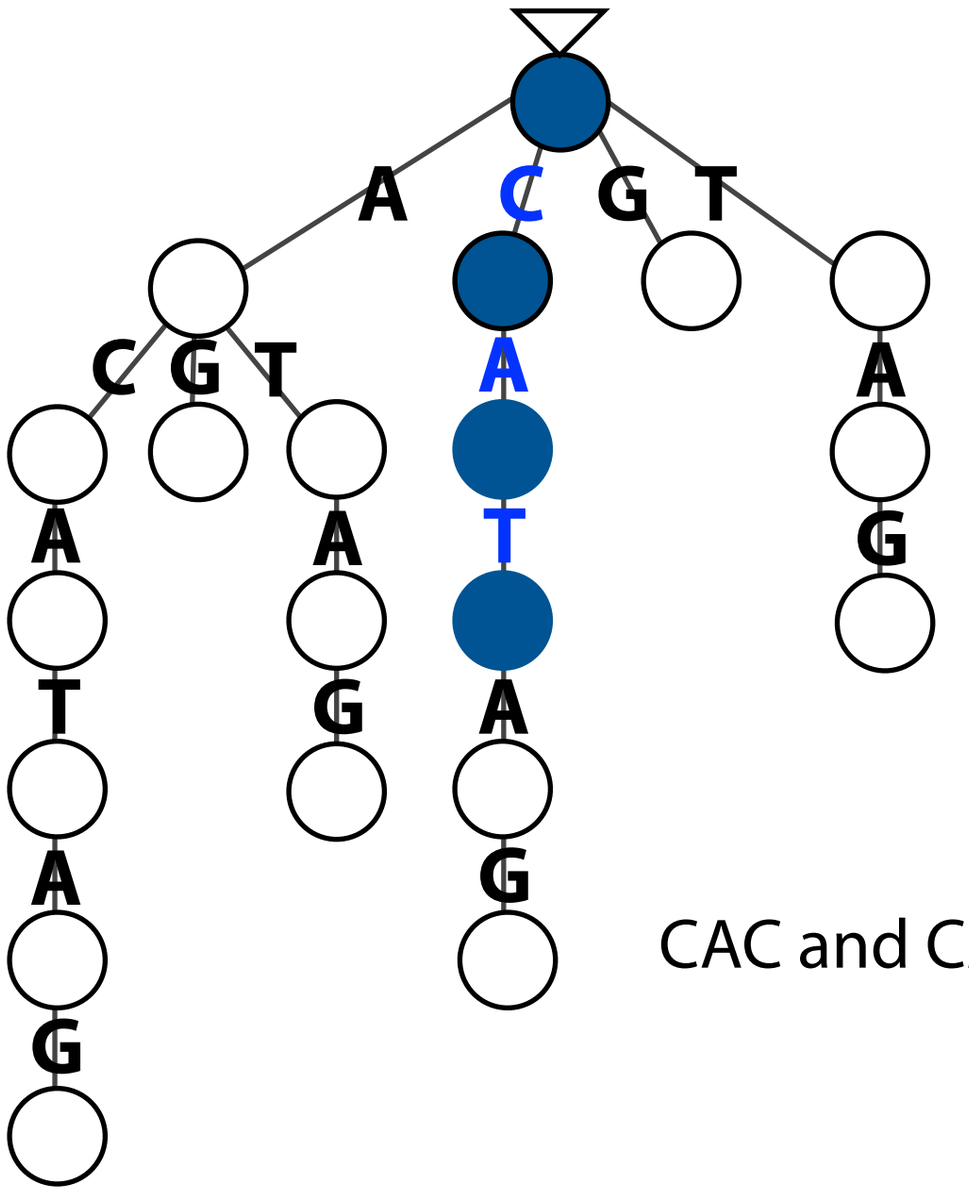


Next try CAA: we fall off suffix trie after CA, before CAA

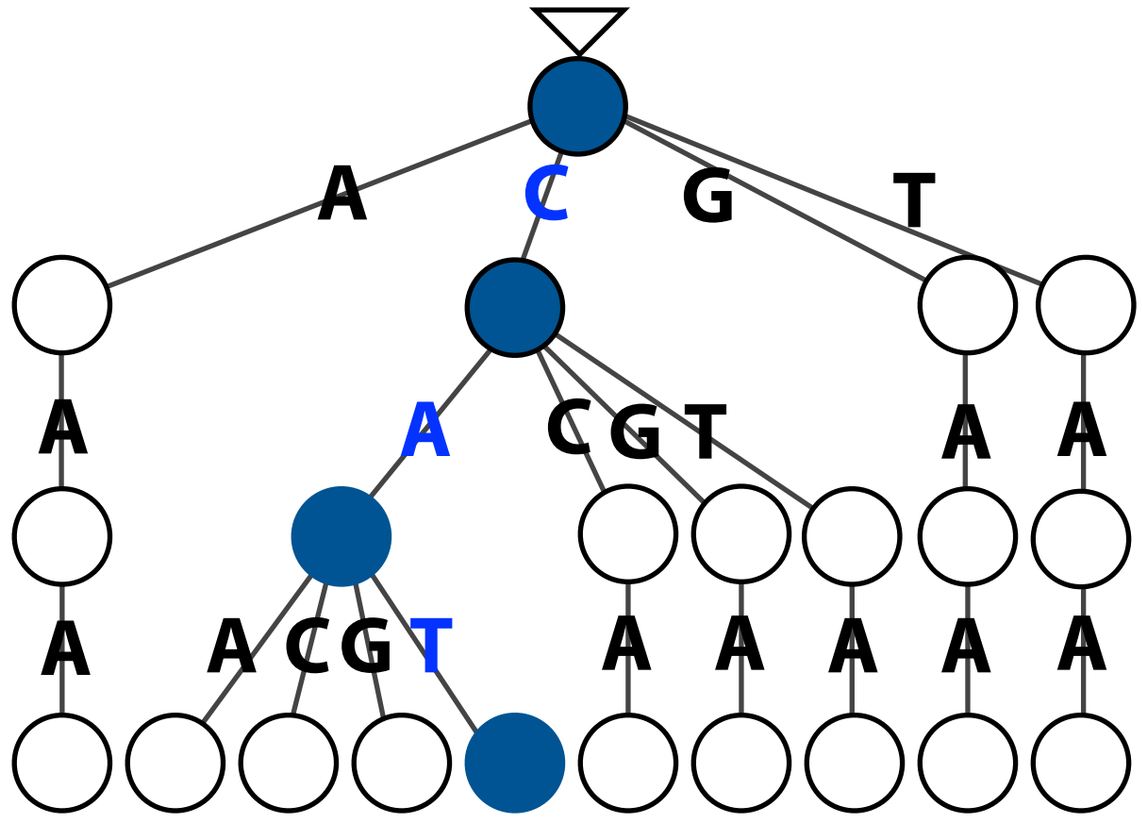
Co-traversal

Say index is a suffix trie. Imagine querying it with each neighbor.

Suffix trie of T = ACATAG



Trie for neighborhood within 1 mismatch of P = CAA

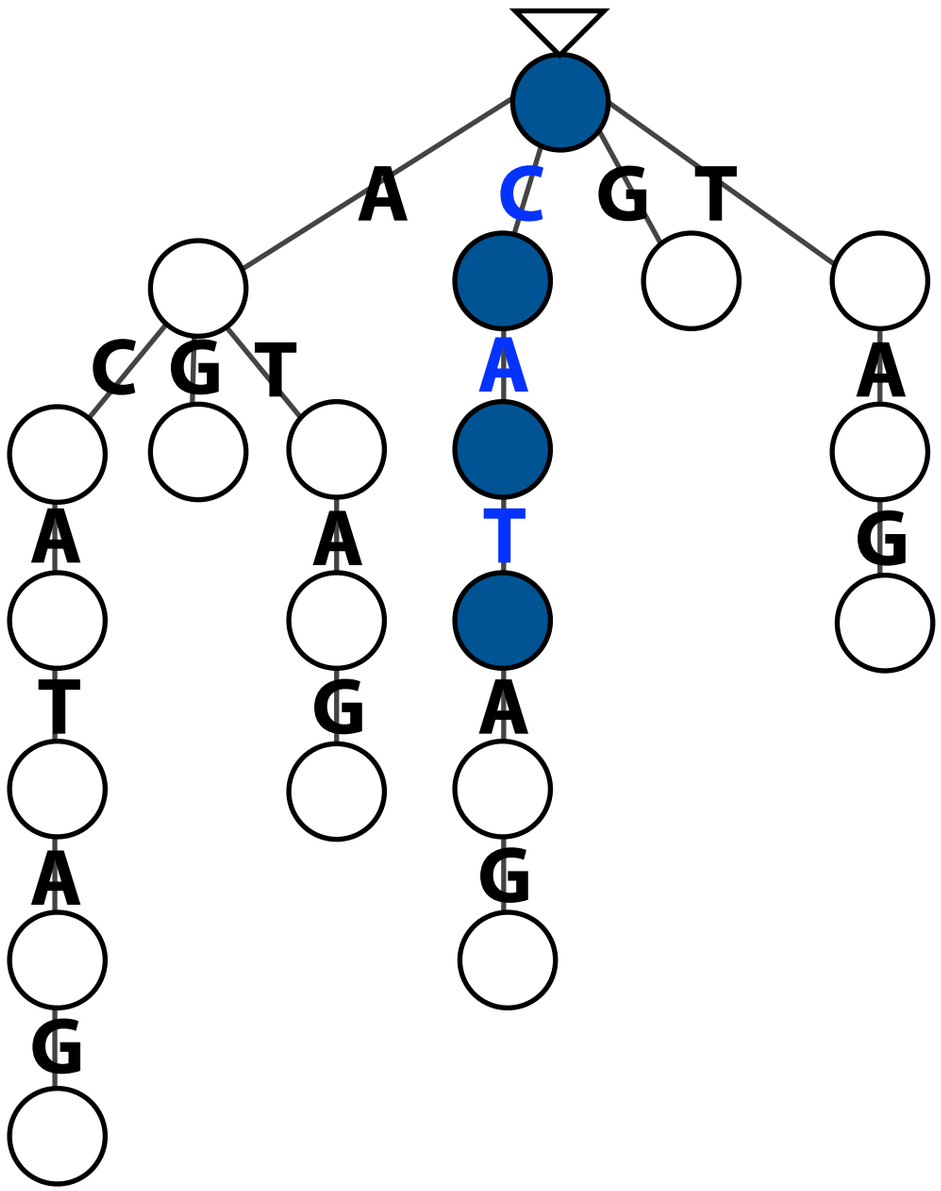


CAC and CAG also fail. Next try CAT: success.

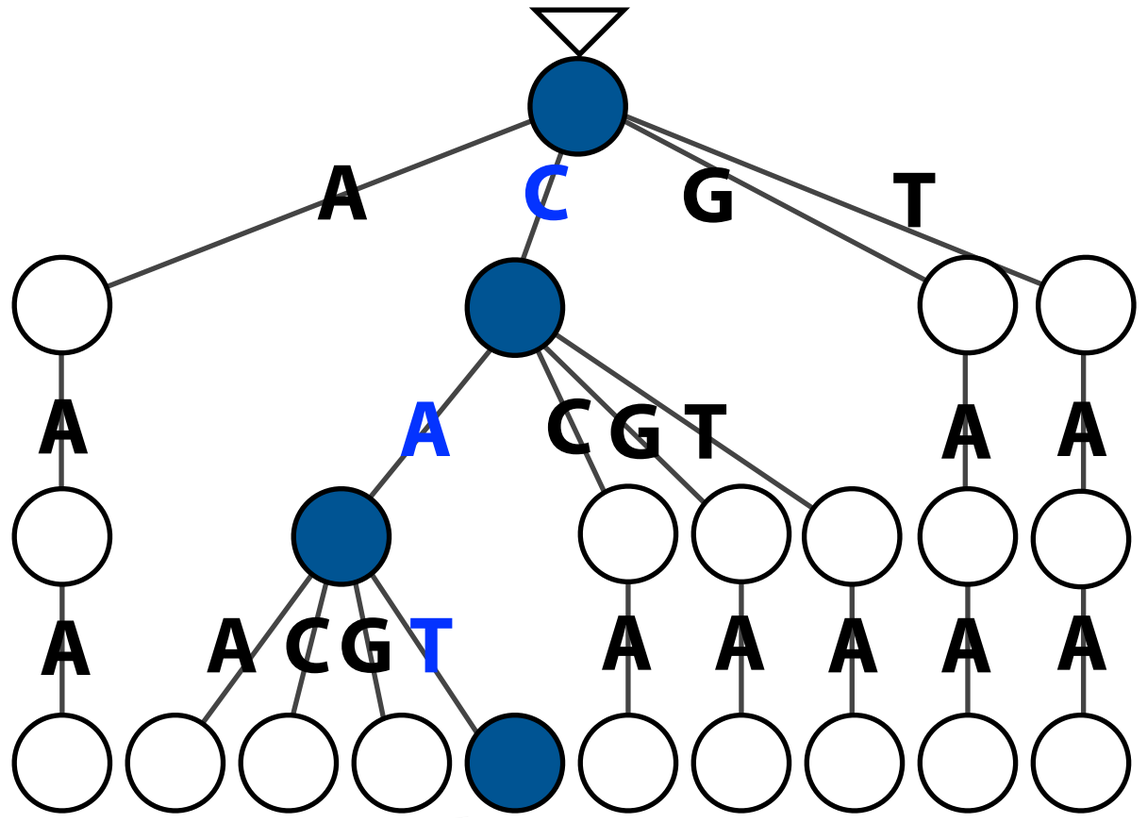
Co-traversal

Say index is a suffix trie. Imagine querying it with each neighbor.

Suffix trie of T = ACATAG



Trie for neighborhood within 1 mismatch of P = CAA



Common path ending in a neighbor leaf corresponds to an alignment of a neighbor of P to a substring of T

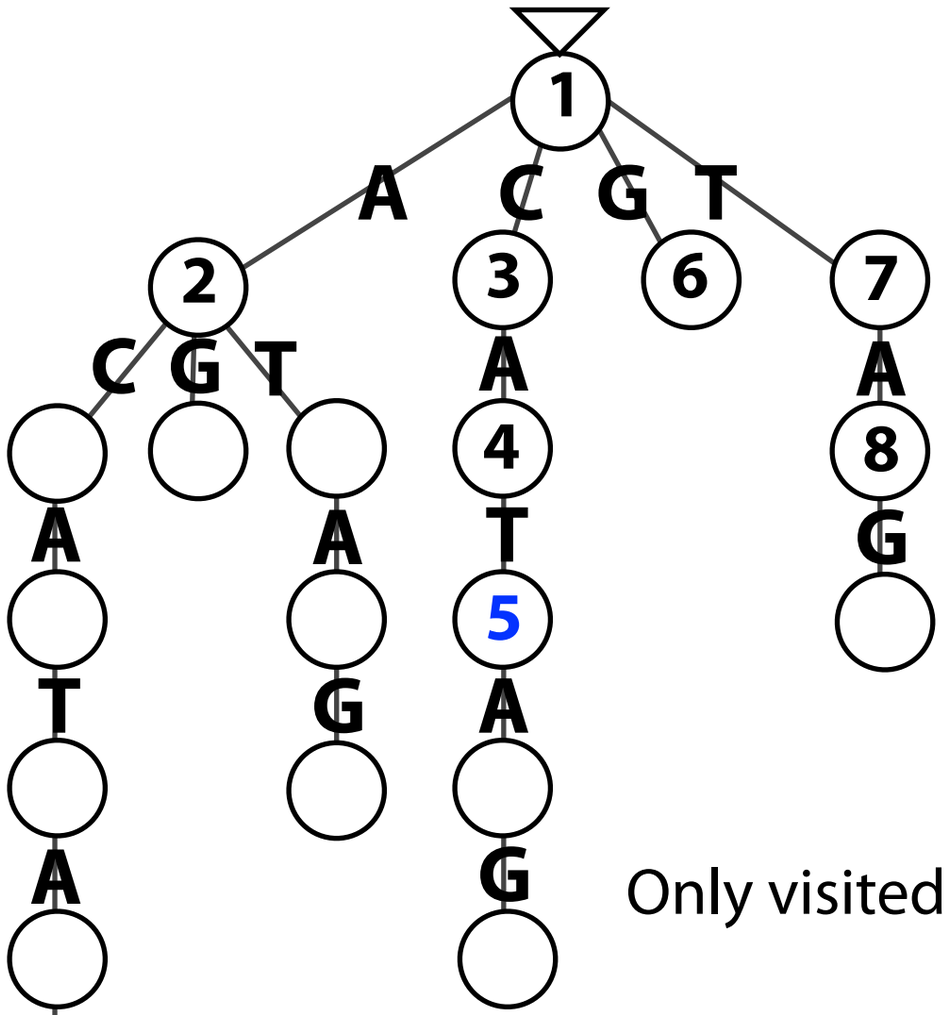
T: A C A T A G

P: C A A

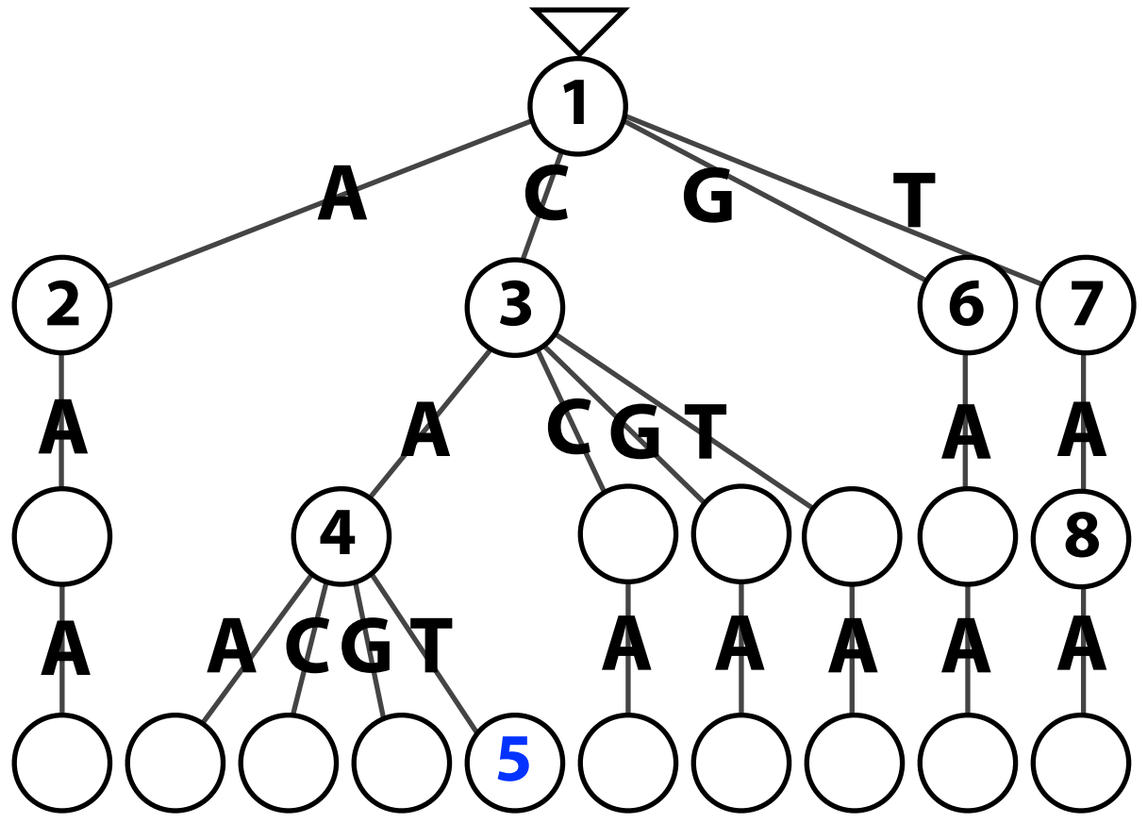
Co-traversal

We can find all such alignments with *co-traversal*: explore all paths that are present in both trees and end in a neighbor leaf

Lexicographical depth-first co-traversal visits node pairs in this order:



Suffix trie of T = ACATAG

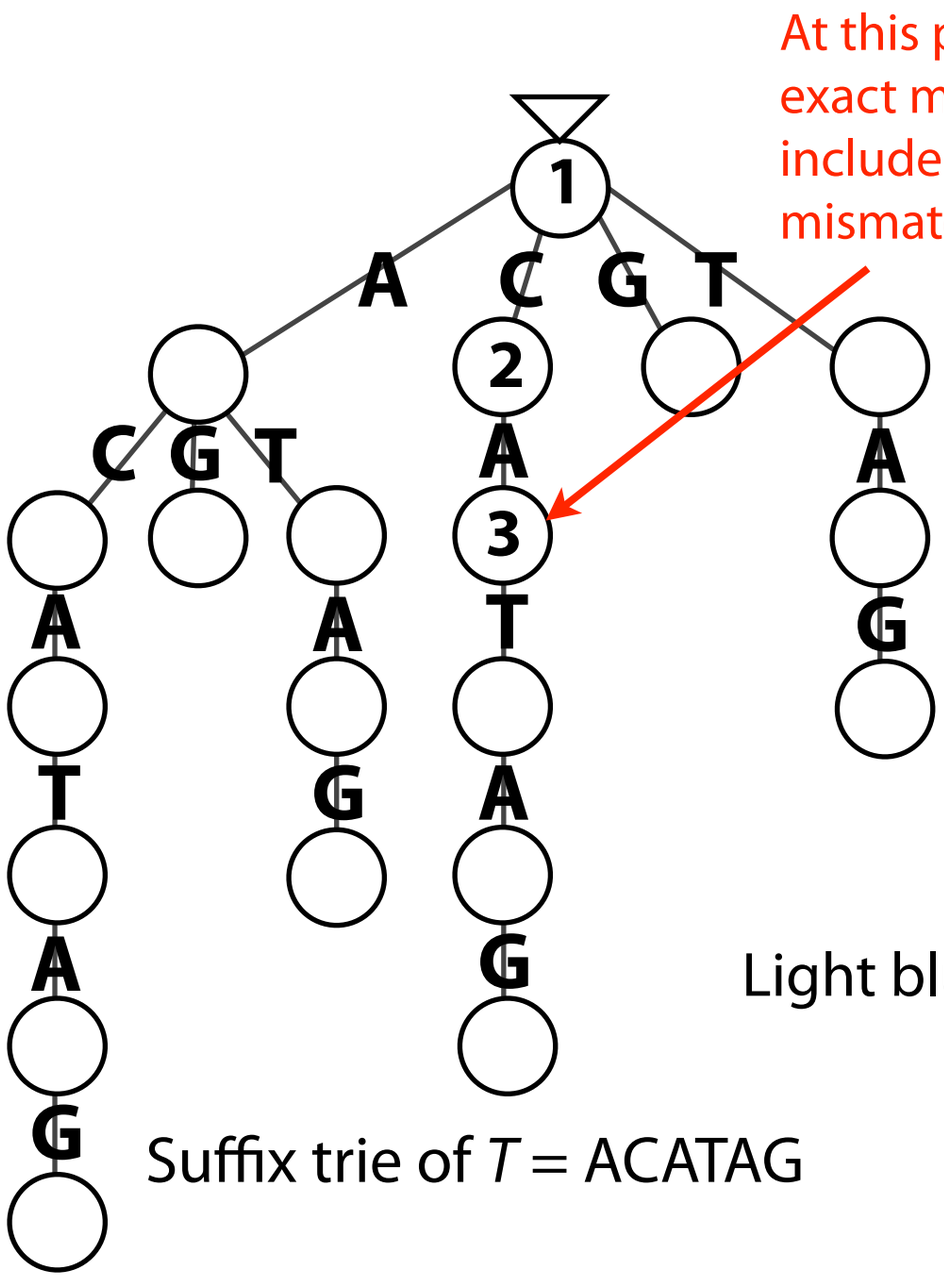


Only visited 8 nodes in these 20- and 22-node tries

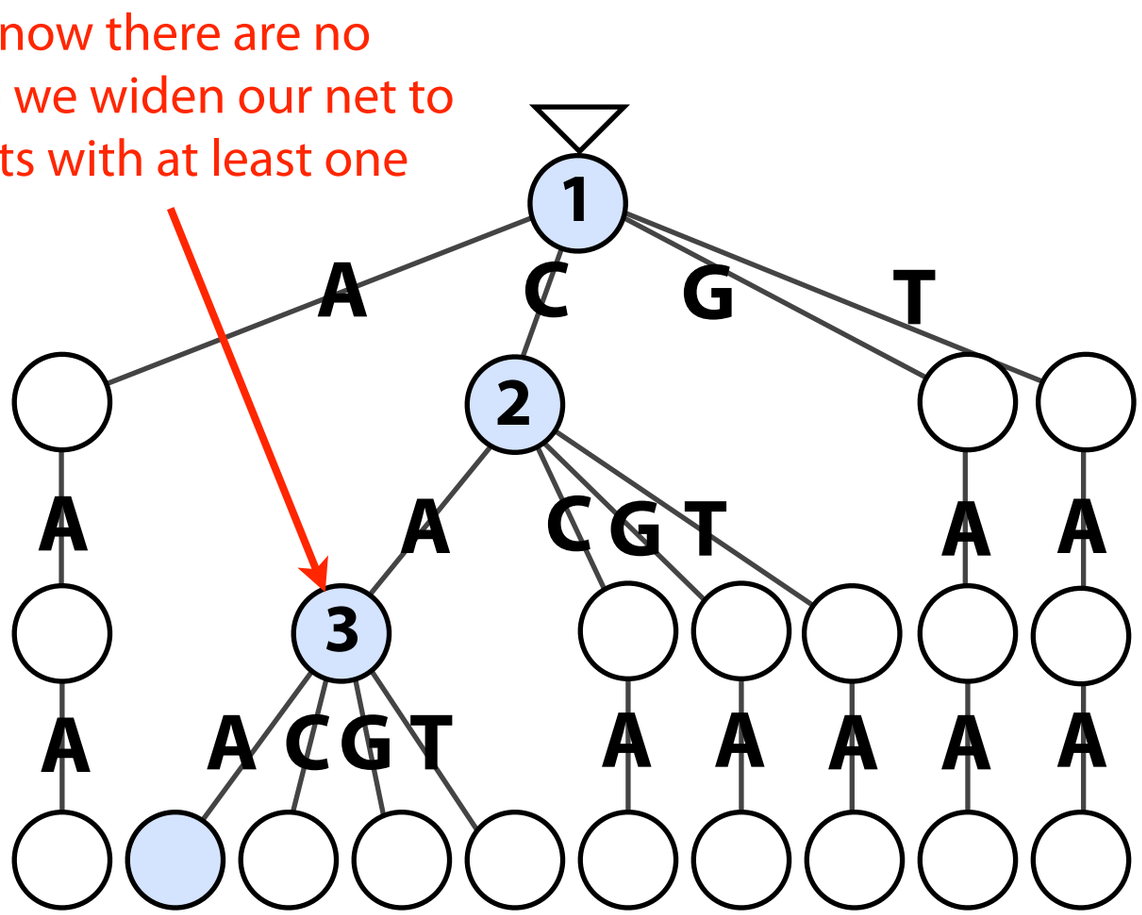
Trie for neighborhood within 1 mismatch of P = CAA

Co-traversal

We can also conduct *best-first* search, visiting paths with fewer edits before paths with more edits:



Suffix trie of T = ACATAG

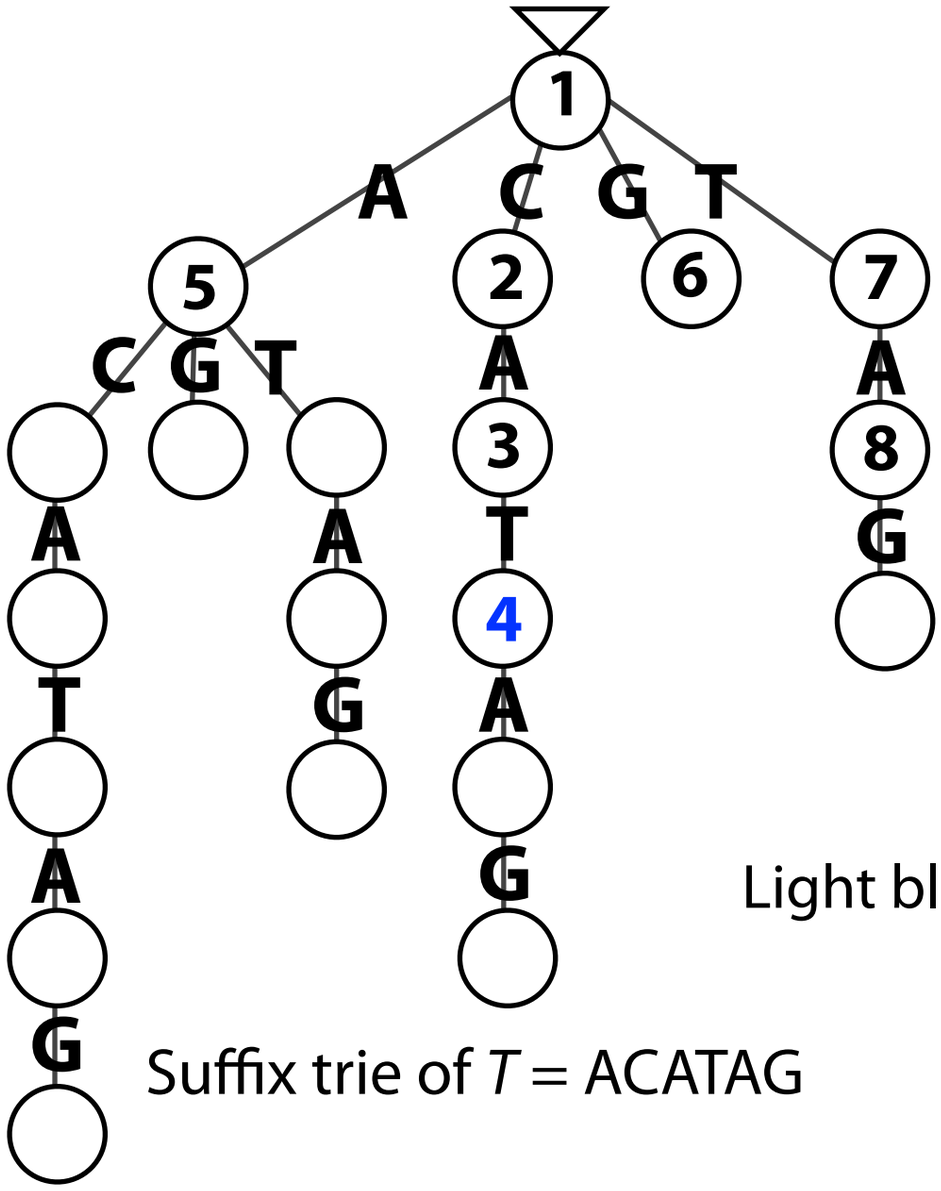


Light blue nodes match P exactly. Others have 1 mismatch.

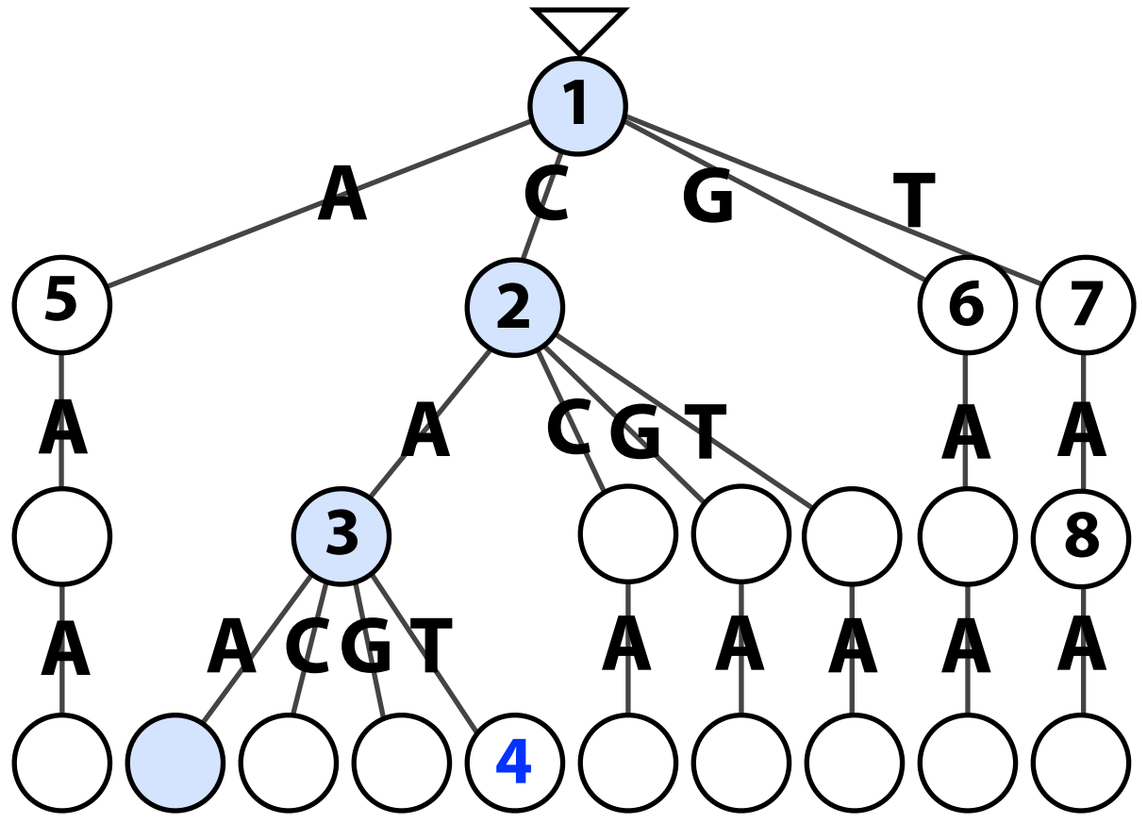
Trie for neighborhood within 1 mismatch of P = CAA

Co-traversal

We can also conduct *best-first* search, visiting paths with fewer edits before paths with more edits:



Suffix trie of T = ACATAG

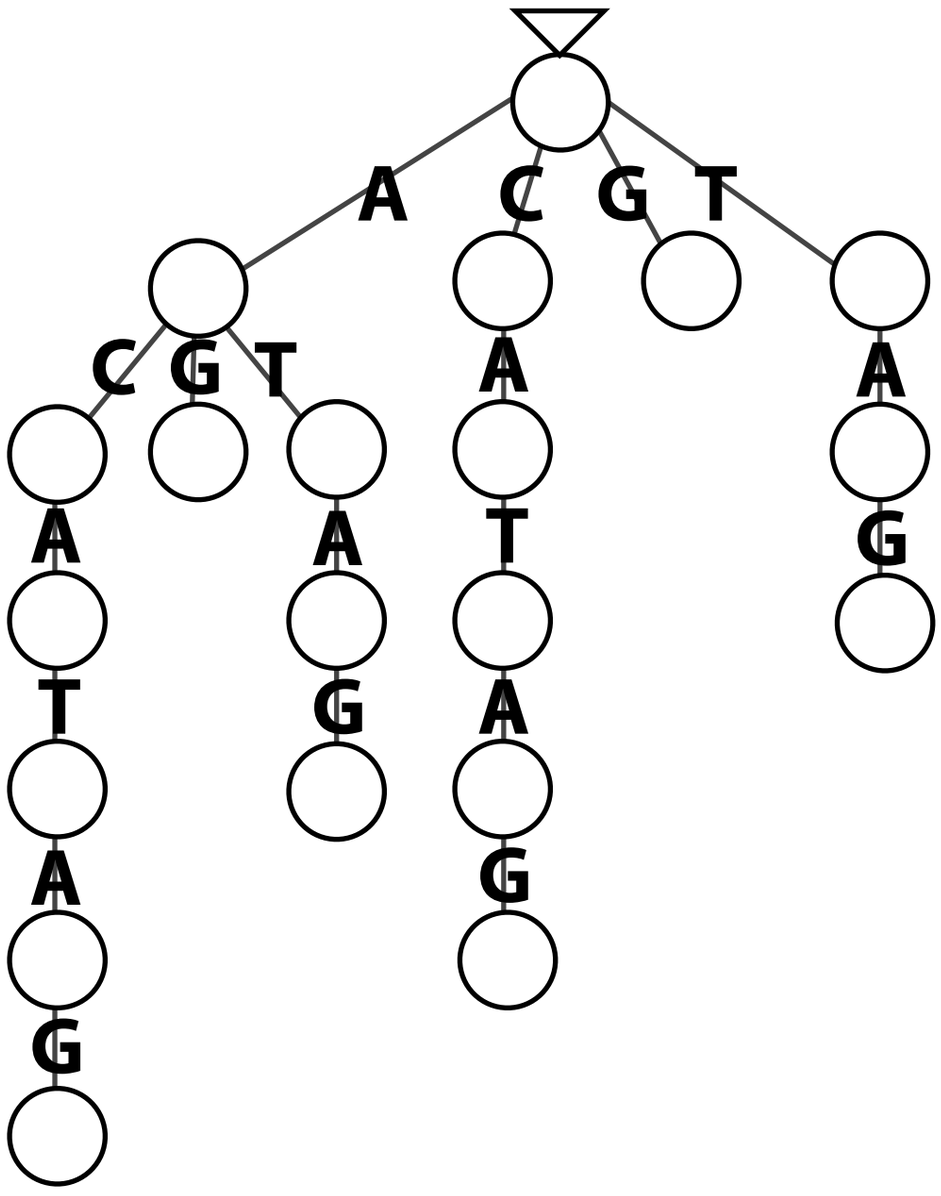


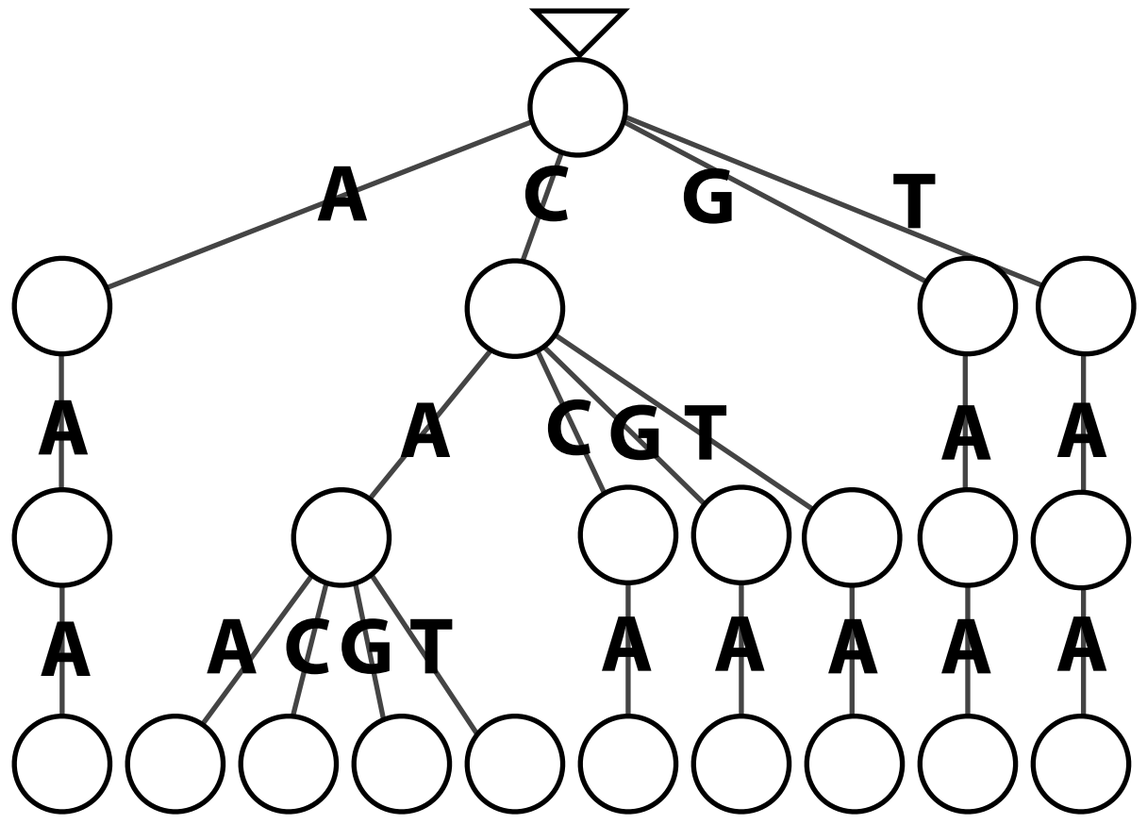
Light blue nodes match P exactly. Others have 1 mismatch.

Trie for neighborhood within 1 mismatch of P = CAA

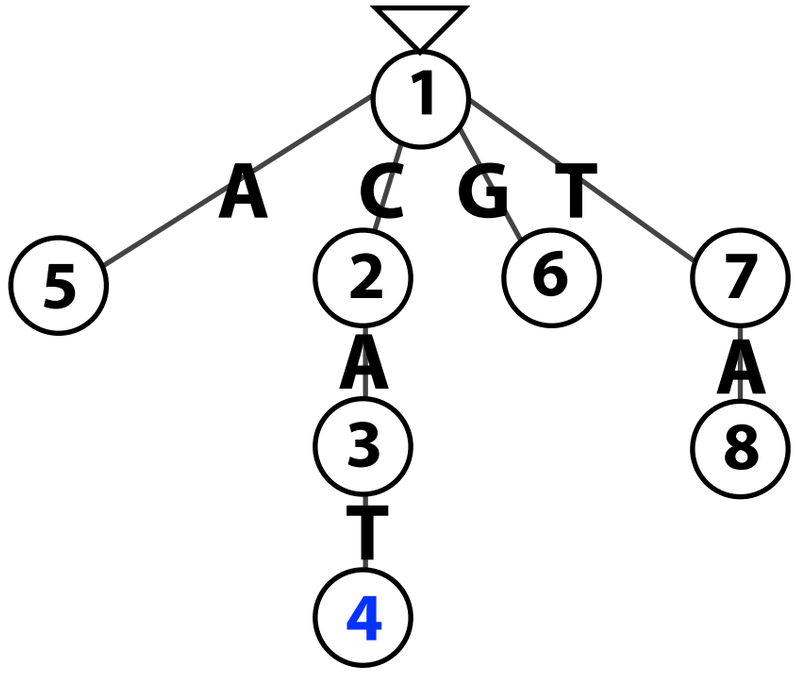
Co-traversal: pruning

We can think of the tree we're exploring as being the *intersection* of these two trees...





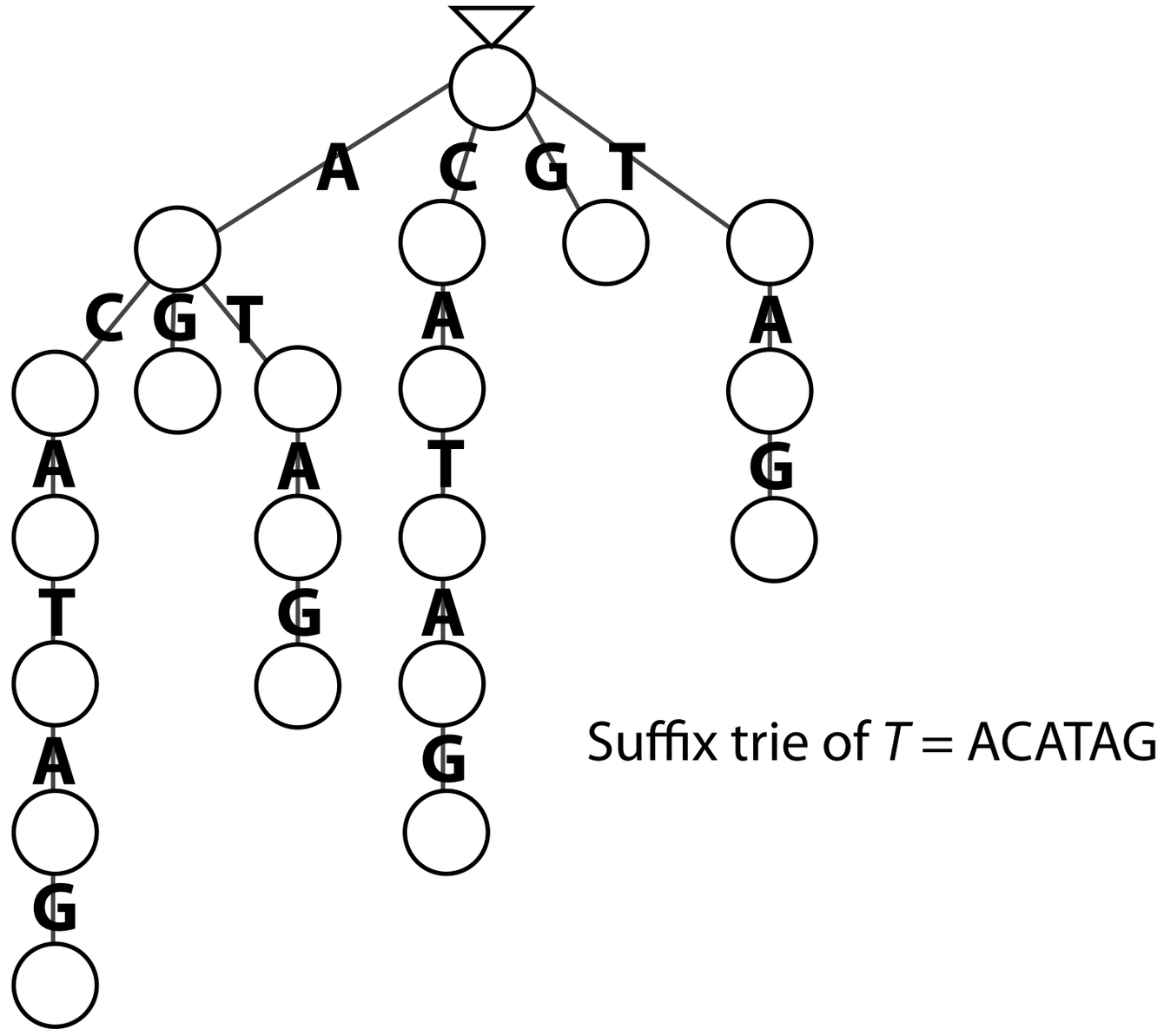
Co-traversal: pruning



Suffix trie of T = ACATAG ∩ Trie for neighborhood within 1 mismatch of P = CAA

Co-traversal: indexing

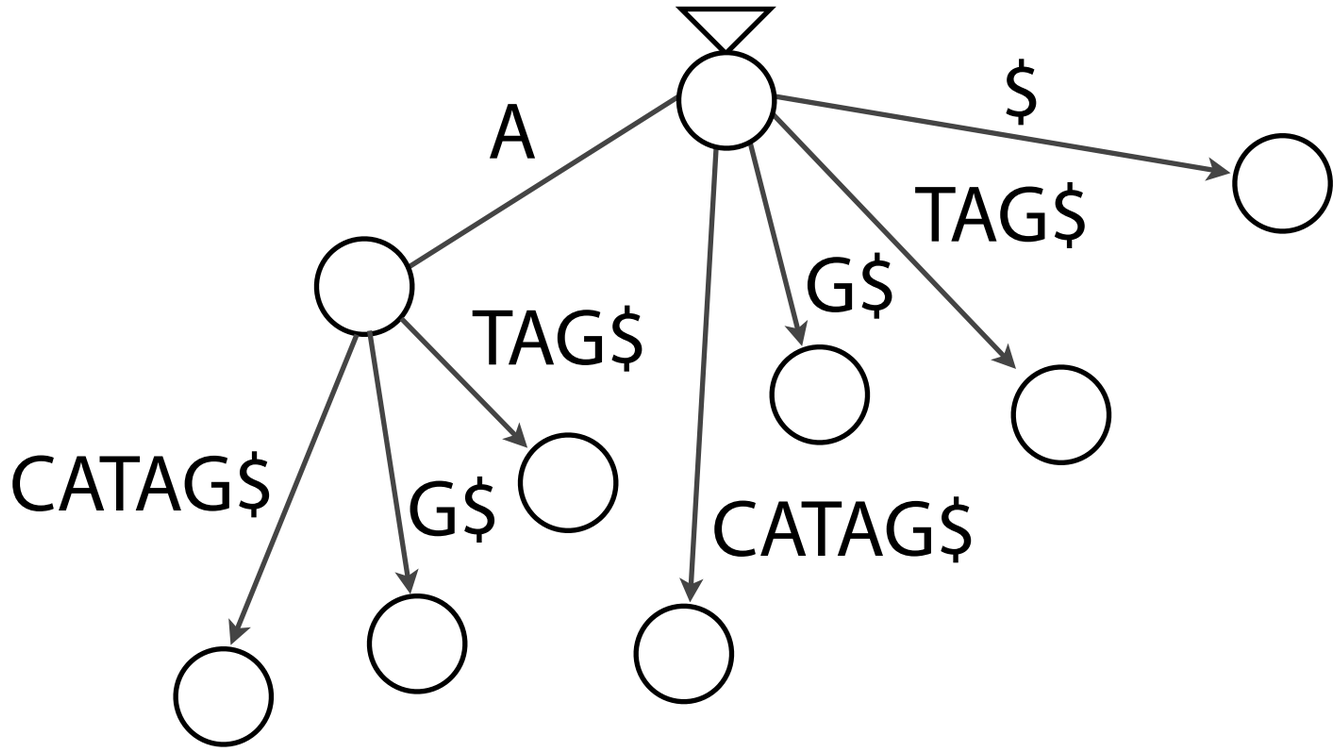
Co-traversal uses the shape of the suffix trie, but we don't want to actually build it. It's $O(m^2)$ space. What's an alternative?



Suffix trie of T = ACATAG

Co-traversal: indexing

Alternative 1: Replace suffix trie with suffix tree



Suffix tree of *T* = ACATAG$

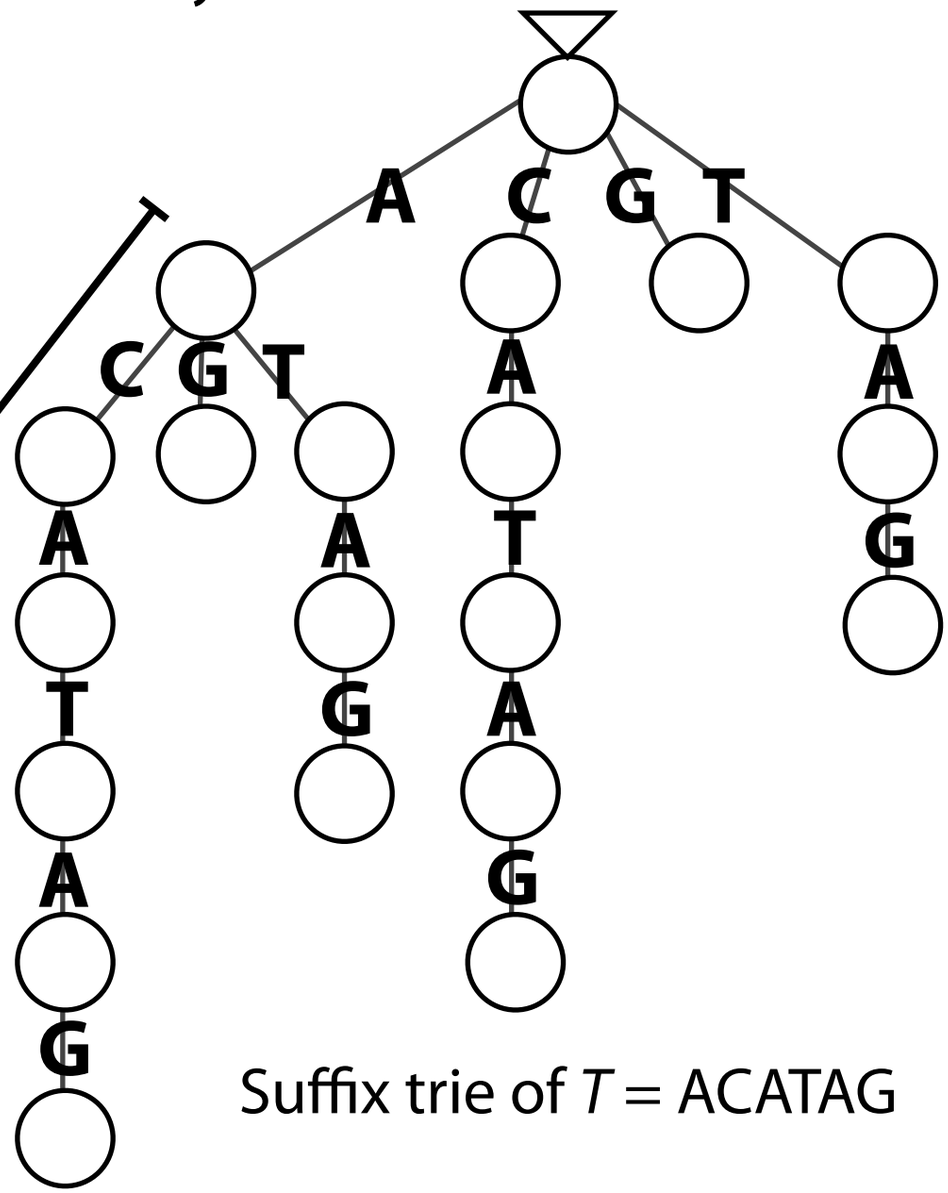
Co-traversal: indexing

Alternative 2: Replace suffix trie with suffix array

6	$
0	A C A T A G $
4	A G $
2	A T A G $
1	C A T A G $
5	G $
3	T A G $

If we know range of SA elements with A as a prefix, additional binary searching gives range with AC as a prefix

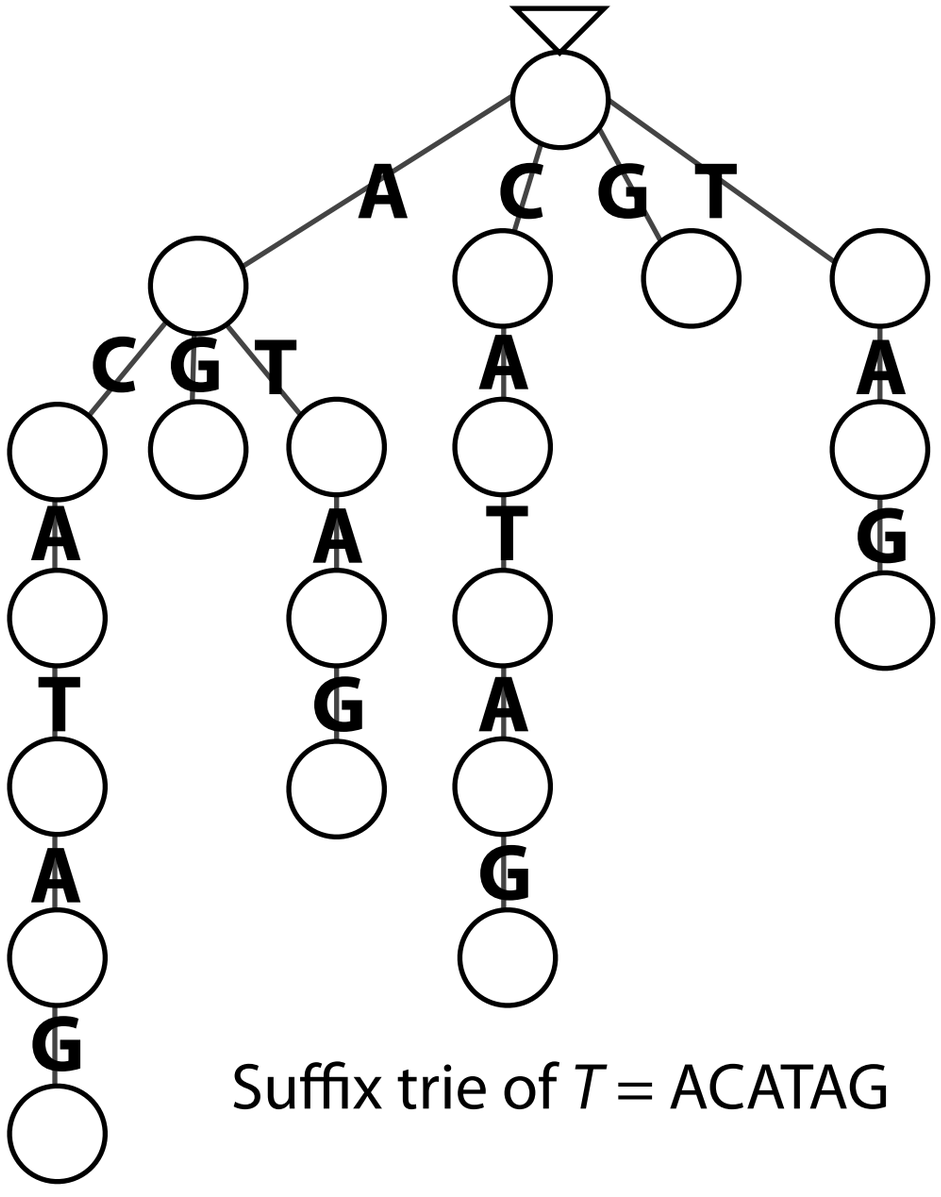
Even if we don't *build* suffix trie/tree, suffix array allows us to *traverse* it



Suffix trie of T = ACATAG

Co-traversal: indexing

Alternative 3: Replace suffix trie with FM Index



Suffix trie of T = ACATAG

Similar argument as suffix array, using LF Mapping instead of binary search

To traverse suffix trie, we need to build FM Index of T' = reverse(T)

Why?

Typical FM Index matches successively longer *suffixes* of P. If we want to match successively longer *prefixes*, we have to reverse T before building FM Index.